

00:00:00,100 --> 00:00:07,440 [speaker_0]

Reported this and posted and be horrible to each other on the internet. I, I, I, I say yes. The rest of you?

00:00:07,720 --> 00:00:08,860 [speaker_1]

Yeah. Yeah.

00:00:08,920 --> 00:04:46,960 [speaker_0]

Appreciate you. Okay. Book, vroom, and zoom. Restoring theoretical physics, mathematics, and science in the US national interest. Let's say start off with some really tough stuff. We... I am not actually an entertainer. I'm really closest to being a physicist. I don't have a PhD in physics, but I have a PhD in mathematics, and I use that to circumvent the problems of string theory in my head. The reason that the outside world cares about us is either they want something for us, or they wanna know who they are and what this place is. We are the custodians of the most important knowledge of the universe. And I don't know how to emphasize that enough. What we do is glorious, and unfortunately, it also, uh, brought the hydrogen bomb in November of nineteen fifty-two. So we are the VCAD line. That was a project between a geometer and a physicist, and the geometer was Stanislaw Ulam. And the physicist was Edward Teller, a truly great man who was also, in many ways, one of our most controversial figures. We are the custodians. We are responsible for bringing the doomsday ability that stops the fourth turning and all that kind of stuff. Um, I, I can't tell you how important we are. We don't feel it, and we don't think about it because we haven't succeeded in those terms in so long. But really, humanity rests, as I see it, on theoretical physics. Sheldon Cooper, I think a mashup of Sheldon Vasha and Leon Cooper of Cooper pair fame, gave up on string theory in twenty fourteen, twelve years ago. That was a reflection of the zeitgeist that this was not the fun, sexy, cool thing anymore. Uh, that was advised, as you know, by a physicist in California. So Sheldon realized this is twenty years of work on string theory, hit, hit a dead end, and decided to switch his research focus to dark matter. That tells you, in some sense, how out of whack the public's perception, our perception of ourselves, and the government's perception of funding can be. So you have to actually take that as indicative and important that we're not reaching the public. We're still seeing these articles about string theory, about Michio Kaku says that dark, you know, dark energy is leaking from extra dimensions. Who knows what? We don't have a good brand and a coherent brand the way we would if things were humming along well. When things are humming along well, everybody knows who's got the ball. There are different factions. We can watch different teams spar, but it's all one coherent story. Right now, we're in a period of incoherence. So what the heck happened to theoretical physics? It is almost impossible to relate to these people. The density of successful human beings in that photo should give us pause. You will notice that very few of them are dark of hue. Very few of them are female. Although, all of the females got two Nobel Prizes. The men can't say that. Um, although I will point out that the one who got a second Nobel Prize wasn't allowed to go to Stockholm because she was carrying on with a married man. So this is the world of slut-shaming in our community. [laughs] My question is, do you see yourselves, those of you in theoret-- Who's in theoretical physics in one form or another in this room? Do we see ourselves as those people? My claim is we are. We don't know it. You know, this is like that weird scene, "I have a gift," you know, and the mysterious stranger comes to town and, "But I don't feel that way." Yeah, that's what we are. We're supposed to relate to these people. Now, when I say, "What happened here?" I had a complete break with reality when I went on Piers Morgan, and let me see if I can get this to play. [video playing] ... situation, uh, is nearly constant for reasons that completely elude me.

00:04:48,360 --> 00:06:12,756 [speaker_2]

Sheldon. The good news is I have read Eric's paper. Here it is. I actually have it here, right here. First thing you gotta do is make sure that your theory makes contact with modern physics as it is understood. If you have a new paper out, physicists are gonna look at it. They're gonna look

for, you know, where's the Lagrangian? [upbeat music] Where's the interactions? [upbeat music] Is the proton stable? [upbeat music] Is there dark matter? Like, how does it fit into what I already know? [upbeat music] Are there any new predictions? [upbeat music] Eric's paper has none of that. Again, none of that is there. And again, I see none of that in Eric's paper. None of that. None of that. None of that.

00:06:13,115 --> 00:18:34,648 [speaker_0]

[upbeat music] [sighs] It's at the top of my Twitter feed, and I believe it has something like, uh, two and a half million impressions or something like that. Why lie? I don't know whether the lie is I read, or I don't know where the lie is it doesn't have any of these things or what is going on. But ultimately, something very bizarre is happening in theoretical physics, where we have these narrative truths that are completely untruth, that are highly durable. And in science, there's a, a role for disconfirmation. I was clearly wrong most of my life about UFOs. I thought there was absolutely nothing there. I thought it was the dumbest thing I'd ever heard. I thought it was a ghost story told around a campfire. I still don't know what's there. I have no proof whatsoever that there are any alien or craft or anything. At a minimum, just so it's called a special access program or a WAP USAP, a Waived Ambiguously Unacknowledged special access program, one or more of these things involving UFOs, perhaps as a cover story for the, the development of stealth aircraft. I don't know. But there's not nothing there. Now, many of us want there to be nothing because it's garbage, it's junky, it's cringe-worthy. Tough shit. There is something going on with our government and our airspace and UFOs, and it is completely unknown. And pretending that it doesn't exist so that you can continue to have this kind of erudite, you know, sip your tea with your pinky in the air, it's not gonna work. Something's happening. We have to recognize that something bad is happening in physics where we're just lying about all sorts of things. It's like saying I do not theory because plasmids are nucleic acids in circles, and therefore, the regulation of gene expression is caught up in lies. But it's not true. But my old professor, Raul Bott, had it on his grant, and I said, "How, how is it that you're keeping plasmids?" He said, "What's a plasmid?" It had been inserted by a secretary in order to justify his grant. We are lying about various things, which is okay, but we're lying too much. So my impossible job is to convince you that you are in a suicide cargo cult, as Richard Feynman once warned us. Now, this is the top of my Twitter feed, and I thought that by putting a Lagrangian, or to be entirely accurate, an action, the integration of a Lagrangian at the top of the feed after Sean Carroll, uh, told me that I don't have a Lagrangian. By the way, he was joined by Steven Hossenfelder, who said, "Eric doesn't have a Lagrangian. He doesn't have any equations." But then again, neither do the rest of them. I have no idea how these minds are working. And I know that I have not been engaged once, despite the fact that I have over a million followers and I've chosen to make-- with all the things I could do to promote myself, not one single person has looked at this thing and recognized that this is the Carroll-Field-Jackiw Lagrangian or action on the bottom. And this is the geometric unity first-order Lagrangian. Above, we have direct one-to-one correspondence with those vertical arrows, and all those zeros are things that Sean did not know how to do and put in. And you'll notice that Sean's term P is a Lorentz-violating term. His Lagrangian is garbage. He's a direct competitor, and he's using his role as tastemaker, as influencer, as gatekeeper, as peer reviewer to say, "My competitor does not exist." I have no idea why. What is the nature of the academic science cult, or cults? We have these gated consensus narratives. So in physics-- I have been in math, physics, and econ departments. It is not known to most people that these same academic issues crop up field to field. So in physics, we have the only game in town, quantum gravity or string theory or M theory. In biology, we have neo-Darwinism, really evolutionary theory. There's something called the central dogma of neurobiology theory. We don't get into that right now. In political science, you might have the Washington Consensus, which would be a quote on the other side of that. In economics, you have neoclassical economics or economic imperialism. These are basically ways of scoping the Overton window of a field, and so things inside that window are considered legitimate academic topics, and things that question the foundations are, in general, left to

languish and die. And the point is to keep out the infection of complicating realities in all cases. This is not working. All sorts of narratives that used to work did not survive the social internet age. Now we have search. Anybody can open a YouTube channel. You had the ability to gate the three networks in the Walter Cronkite, Eric Severeid days that none of you remember. It seems like maybe one of us remember. Um, that doesn't work anymore. We have to adapt to the fact that people are going to raise issues, and if what we choose to do is to say that legitimate person is illegitimate, ultimately, that'll work for a while, but it's going to come back, and it is going to blow up on us, and it is blowing up on us now in the current moment. You lost the support of tech by staying silent. How many of you spent significant time in Silicon Valley hanging around with billionaires? You have no idea what happened. Somehow, and I don't really understand it, Fauci became the emblem of all science, and in particular university science. And what our elite figured is that instead of contributing to universities and funding prizes and revering scientists, they somehow got this idea in their head, "We've been had. We've been had because somehow the narrative isn't right. Why is it I can't ask, might it be the Wuhan Institute of Virology and its Biosafety Level Four laboratory or even Biosafety Level Two laboratory that might be at the heart of the COVID origins? Why can I not ask questions about Ralph Baric in his lab at the University of North Carolina Chapel Hill? Why is Peter Daszak, a zoologist from Ukraine, from a Ukrainian background, able to get so much money from DTRA, the Defense Threat Reduction Agency, uh, in a nonprofit that seems to be studying almost exactly the thing that this virus is? Why are there twelve nucleotides and four amino acids constituting a furin cleavage site occurring directly in spike protein? Is this some sort of mysterious zoonotic origin?" It's not the fact that we know or we don't know. I don't know the answers, but I can tell you this. Everybody in Silicon Valley felt the idea that I was not... That, that they were not allowed to ask questions. And the fact that they weren't allowed to ask questions meant that they checked their cartoon concept of science and they said, "I thought science was the thing that said we could ask questions without being called a racist, without being called a troglodyte, an anti-vaxxer, a conspiracy theorist, a charlatan, a grifter." All of this crap that is now infecting your minds, and I want you to think about the sandbox in your mind which you scroll and how many of you have heard, have heard the bizarre terms grifter and charlatan, which are extremely rare in speech but common on the internet? This is malware. And I'm not telling you to load my malware into your brain. I'm saying just note it metacognitively, that you're learning to absorb things that make absolutely no sense, and with direct confirming evidence, you don't believe your own eyes and you believe your social organizations. So there's been a catastrophic loss of support from tech. Many of you are unaware of this. Fauci, and to a lesser extent, Francis Collins, Ralph Baric, and Peter Daszak became the face of lying coin-operated science in the eyes of the tech tether. You will also note that the role of public intellectuals, which many of you are supposed to be, has been subsumed as one part of the portfolio of Marc Andreessen, Naval Ravikant, Balaji Srinivasan, Peter Thiel, Elon Musk. Gradually, you are being supplanted by people who at least know how to make money. This is incredibly dangerous. I love many of these people. But it is not responsible to have these people who talk their book, which is part of what they do for a living, as the unbiased source of reason. They need to be counterbalanced by a strong scientific community, and when I tell them this, they tell me, "Eric, science is over. Universities are over. Get over yourself." [audience chuckling] Result: Thirteen PCAST members are announced, Presidential Council of Advisors on Science and Technology. I want you to look at this list and marvel, all right? This is something that the presidential science advisor used to be, uh, Rabi, Isidor Rabi, the guy who said, "Who, who ordered that?" when the muon showed up, heralding the second generation of particle... It is now a guy named Michael Kratsios, a friend of mine, who used to work across Peter Thiel's office from me. He was Peter Thiel's chief of staff. I think he was a, a Yale humanities major. I never had a scientific conversation with him. Not- nothing against Michael. They don't know any scientists. How many scientists in this room have dined at Mar-a-Lago? Raise your hand. [audience chuckling] If you're not dining at Mar-a-Lago, you're not part of the conversation. And you notice I did not raise my hand. I am not going to Mar-a-Lago. [phone dinging] None of us are present in these stor- in these rooms,

and we cannot afford not to be present. I don't care how much you love or hate these people. Course correction from the scientific community is absolutely essential to the well-functioning of the United States and its role as the umbrella of security in the world. Now, I am not starry-eyed about the United States. We do a lot of bad things. But I believe that net we are an on force... on balance, a force for good in the world, and it's only on balance, and that means we have to up the good that we do to counterbalance the other things that we do in matters of national interest. And if that doesn't come from the scientific community, a community that we try to keep the purest, not pure, but purer than everything else, we have no ability to come to ground truth. When we allow our politics to infect what we, what we see, we are no longer capable of being the arbiters of decision. How many saw me talk to Terrence Howard on the Joe Rogan program when Terrence asserted that one times one equals two, and I was getting questions from all sorts of grown-ups, "Is this true?" Okay, we're talking about the catastrophic loss of communal certainty about ground truth. And if we don't establish that, we don't take that not only as our right, but as our duty, every possible thing that can be asserted will be asserted by someone, and we will have a, "I don't know," response from our policy.

00:18:35,548 --> 00:18:38,058 [speaker_3]

Can I ask a question? Why is that name asterisked?

00:18:38,428 --> 00:18:38,707 [speaker_4]

Yeah.

00:18:39,448 --> 00:18:40,288 [speaker_0]

Because that is-- this is-

00:18:40,868 --> 00:18:41,088 [speaker_4]

Oh.

00:18:41,308 --> 00:19:34,808 [speaker_0]

That's the only actual scientist that I recognize on that list. May, maybe that... Really great question. That is how bad the situation is. By the way, when I, I let off some nuts and bolts stuff. Many of you see me as a tech guy or many of you guys just see me wrong, and I can't do anything about that. Um, I got a call from David Friedberg immediately after I let off some stuff. All of the people on the All-In podcast called me. And he said, "Well, what, what do you want us to do?" And what I wanted to say was, "Talk to my friends, the physicists, who hate me. Talk to my friend, the physicists, who lie about my work. Talk to the friend, my friend, the physicist, who haven't had-- bought a base hit since, uh, nineteen eighty-three or earlier in a certain sense." And I couldn't.

00:19:36,788 --> 00:19:36,798 [speaker_4]

Why?

00:19:36,798 --> 00:22:09,448 [speaker_0]

I couldn't because the community has to improve before I can sell it. I'm, I'm, I'm, I'm not going to be a salesman for a product that I-- I absolutely believe in theoretical physics. It is the bedrock of my life. And this instantiation of the theoretical physics community is not up to standard. We've allowed our standards to fall, to plummet, to crash through the floorboards in forty-two years in a way that I had never imagined. And if you go back, and this is an exercise, go back to nineteen eighty-three and before and look what people were doing. It is an unrecognizable field. And if you want to try this with the LLMs, ask it for evaluations of ideas in physics saying, "You're only allowed to use, uh, the corpus before nineteen eighty-- uh, eighty-three and before," or, "You're allowed to use the full corpus." And very often it will tell you, "This is a lousy paper if I use the full corpus, and it's an excellent paper if I use the

eighty-three and before." There's something fundamentally that changed in nineteen eighty-three, eighty-four that is night and day different. And if you don't know this, you have to go back to older papers and books in order to see it. Has quantum gravity always been the so-called holy grail of fundamental physics? There's a lot of narrative support for this. "Quantum gravity is the holy grail," James Gates. [chuckles] "The challenges all center on the question of what is the quantum theory of gravity?" Gary Gibbons. "Quantum gravity is the holy grail," the great Stephen Hawking. "Quantum gravity is the holy grail of theoretical physics," a much debated statement, possibly. This graph, uh, I don't think anyone else has produced it. I took the age of the youngest living Nobel laureate in pure theory of fundamental physics over time. So in other words, we start off with Niels Bohr, go to de-- Louis de Broglie. We get to Werner Heisenberg and Paul Dirac. We don't get anyone young then. Yukawa in Japan, C.D. Lee, uh, with the weak force, Murray Gell-Mann and his quark madness, Steven Weinberg. And then for the first time, we don't have any theorists. I'm wondering, was it something I said?

00:22:10,248 --> 00:22:10,458 [speaker_4]

[laughs]

00:22:10,768 --> 00:26:13,636 [speaker_0]

Um, we don't have any theorists really after nineteen seventy-three, uh, in terms of work. Frank Wilczek and Ed Witten were both born in nineteen fifty-one. Frank Wilczek is slightly older, and Frank Wilczek is sort of the dividing line, the last young person to make contact with physical reality in this particular fashion. So what happens is now that our, our youngest theoretical physicist who has made a fundamental contribution is seventy-five years old now. So we don't even dream about the Nobel Prizes. We start dreaming about other people's prizes from other games. "Hey, what about Clara Ingrid? What do you think of the new Gold Street?" As if that matters to any of us or should matter to any of us, because we've lost our own, our own prizes like academic freedom or the privilege of actually making physical progress. One of the most subversive instruments that you're allowed to have access to is Google Ngram. You ask the question, how ancient is peer review? Well, of course, peer review goes back to the founding of the Royal Society. It's an absolutely essential part, the sine qua non of scientific excellence. This is nonsense. This is a narrative that mostly came from Robert Merton. It was retconned in order to explain some things that happened in nineteen sixty-five and nineteen seventy-five. What happened was that in nineteen sixty-five, Congress passed the Medicare Act. That meant the US taxpayer became a giant funder of medical procedures. They wanted to know, why are we doing all these medical procedures? The doctors did not want lawmakers poking around asking these questions. They found a clinic in Utah that was using something called peer review. And what they did is they said, "Okay, we doctors will make these determinations. We will be the checks and balances on this." Let the, uh, the heathen stay on the sidelines. We've got this. By the way, that's what the SEC tried to do with mortgage-backed securities and becoming its own self, self-police in Washington. In nineteen seventy-five, unfortunately, the National Science Foundation got involved with cultural anthropology and put out something called MACOS, Man, A Course of Study. Fiscal and cultural conservatives converged on this. Fiscal conservatives said, "Why are we paying for this, this national program to educate the youth? We don't necessarily agree with the content. Why is the National Science Foundation doing this?" Cultural conservatives said, "Wow, you've just relativized Judeo-Christian values, and you're studying all these different groups of people, and what you're saying is, is that they're all equivalent." And what did the NSF do? It said, "Well, how, how do we keep conservative lawmakers from policing our grant-making structures?" And they said, "We'll borrow from the biomedical literature." The, uh, double helix paper in Nature was never peer-reviewed. Einstein, when he was peer-reviewed, was outraged. Peer review was not part of our tradition when we were kicking anise. Your head is filled with things that are not true. And by the way, feel free. You have Internet connections. You can look this up, and you will find what I am reading. And you-- then you can choose to argue. Many people, even when they get

to the truth, they will continue because to say that this is true is to mark yourself as somebody who is not an academic, who doesn't get it. And my claim is the entire appeal of going into academics as I did was I get to stand one against the many. If it's ten thousand of you and I'm right and you're wrong, you better believe I'm going to tell you, you're all wrong and I'm right. And the ability to do that here and to adjudicate and to win is the appeal of theoretical physics.

00:26:16,096 --> 00:26:20,896 [speaker_1]

Does anything good come out of like a Google Labs, or that's all just like good company?

00:26:22,836 --> 00:26:26,956 [speaker_0]

Well, first of all, Google employs people like Adam Brown in quantum gravity.

00:26:27,565 --> 00:26:27,776 [speaker_1]

Yeah.

00:26:28,276 --> 00:26:31,316 [speaker_0]

Um, there's lots of great things that come out of Google.

00:26:31,556 --> 00:26:32,746 [speaker_1]

Okay. Those kinds of things.

00:26:32,836 --> 00:36:36,970 [speaker_0]

Google's also the source of the conspiracy to slander me. Don't understand why. It's a defense contractor. Narrative engineering and, uh, covert, uh, what is it? Covert social operations and so on is a real thing. We have to control narratives. By the way, we, we can't get out of this. We need narratives. So even though I'm going to tell you a lot about bad narratives, it's not like we can function without. Man is a narrative creature. What I'm complaining about is that the narratives are so childish. They're so anti-intellectual. They're so low IQ. They're so D-class A that frankly, we shouldn't have anything to do with them. Right? We're, we're, we're-- Th-this is like an intellectual crack house. What I'm trying to do is to, is to exhort you to raise your game and to embrace higher quality load-bearing fiction. Um, so most of the academics have no real idea of their own history. Peer review comes, as I said, from nineteen sixty-five, the Medicare Act. You can see it. There was no history of quantum gravity, really, um, i-in a, in a s- in the sense that we know it now before Bryce DeWitt was messing around at Tata Institute of Fundamental Research in nineteen fifty-two, whereas India's response to the hydrogen bomb effort was to make sure that it had a center of excellence in South Bombay, in Indian Nagar. It's the most beautiful place in Bombay. It's not the Willingdon Club. It is actually TIFR. I exhort you all to go. It's a wonderful place. Um, Bryce DeWitt went there, had his kid. It was very early, and he was the, really the lone locomotive of quantum gravity up until the early seventies when you had that explosion that I showed you with the worm grains. Um, faculty immigration. If I say to you, should we allow the immigration services of the United States to make our decisions? Many of you will say, "No, science knows no borders. Science is international." This is a big problem because there's another narrative that says, "Give us your US taxpayer money." And we have to have a better response. It is very important to us that the quality of your thinking has nothing to do with your skin, your creed, your country of origin. But I want to show you what actually happened. We had a very high level of faculty immigration, um, where we granted six thousand visas during the expansion era of the universities from nineteen sixty-one to nineteen seventy-six. Remember that we used to educate, um, seven or eight percent of the population pre-war at a post-secondary level. By the early seventies, it became approximately fifty percent or over. So there was a giant period of expansion where each professor could have several professors as students who could then have professors, and then that came to a crashing halt because, you know, if you take, uh, Norman Steinbach, for example, twenty-three students, twenty-two of them became professors who also had doctoral students. Try raising

twenty-two to higher and higher powers and see what happens. Um, in part, that's what happened is, is that we now expect that we will go from something the biologists call K selection, where you have very few ch-ch-children, and you raise them really well and put a lot of parental investment, to r selection, where you have lots of kids, and you expect almost all of them to die. What did that look like after s- the NSF in the nineteen eighties started a panic out of something called the Policy Research and Analysis Division? Under Erich Bloch, there was a claim that there was going to be a massive shortfall or shortage of scientists and engineers based on the demographics of baby boom, baby bust, baby bust being Gen X, the forgotten generation that nobody remembers exists. Um, here's four years in the early nineteen nineties in response to what's called IMMACT ninety or the Immigration Act of nineteen ninety passed in response to the NSF, uh, panic. So more or less what we did is we flooded our market. And the reason we flooded our market was that Erich Bloch charged Peter House, the director of the Policy Research and Analysis Group, and his economist, a person you've never heard of, named Miles Boylan. Miles Boylan, who was a economist that came out of Case Western Reserve, did a secret nineteen eighty-six study in which he said you could depress the wages, uh, of scientists and engineers, which was desirable. The pessimistic scenario was our wages were set to go to over one hundred K by the year two thousand estimated in nineteen eighty-six dollars. And the scientific employers didn't want that. So the NSF went after scientists. You guys don't know that these games are taking place in Washington because you're not invited. How many of you people have spent two weeks or more in Washington, D.C. doing science policy? You are being represented by people who would chill your blood and curl your toes if you knew what they were saying on your behalf. You have no clue how you're represented. COSPA, that is the holy of holy committees inside the National Academy where you put all the heavyweights in case you want to get something done from the government, said, "The increase in science and engineering immigration starting in nineteen ninety-two probably resulted from the Immigration Act of nineteen ninety, which was passed in response to predictions by NSF and others in the late nineteen eighties that a shortage of scientists and engineers was impending." Now let me explain what actually happened. So I moonlight as an economist. As you know, there's a diagram, oh, that looks like the supply curve and the demand curve, where this is the price on the Y-axis. This is the quantity of labor. If you have a supply and a demand curve, all markets clear through a wage mechanism. So I can say I have a shortage of Lamborghinis in my driveway. I have a shortage of Steinways or Bosendorfers, both would be great. I don't have a Bosendorfer grand in my house. Could I afford one? Probably. Why is there a shortage of Bosendorfers? Because I don't feel like paying for it. I have a perfectly fine Kawai upright that's good enough for my purposes. There is no such thing as a labor shortage in a large market economy, full stop, long term. It just doesn't exist. I will clean your toilet with my toothbrush if the price is right. We don't have labor shortages. What they did was they did a secret study where they had both supply and demand, and then they subtracted-- It was very competent, done by Miles Boylan. Then they did a job on it where they moved the demand curves out. So now all they had is supply. They said, "No, no, it's not an economics study. It's a demographics study." And they changed the word shortage to shortfall, panicked all of Washington that we were going to become defenseless, and pushed through this bill, which in the secret study says is meant to depress our wages. That is the design of the study. It's not me being a xenophobe. Okay? I was the guy who found that out, figured out that there had to be a secret study. I went through thousands of pages of microfiche. I finally find this buried in a haystack turned over to the Congressional committee. It's unauthored. There's no author's name. There's no date because they didn't want it traced. I call up NSF, and I say, "Here's the name of the study. I need to speak to its author." I get pushed through three people. Guy picks up on the other end. "This is Miles Boylan." I said, "Miles, did you author the following study?" Very long pause. He says, "I always knew this day would come." You have no idea of your history or why you're here or how this functions. So you were kept as children. The internet trolls are rotting your brain. You were invited here because you are rich and famous, right? No. If that's why I'm here, I don't need to be here. I'm here because you're about to die. Your, your field is in such danger

because you cannot defend yourselves, because you're not-- you haven't elected leaders. You haven't convened groups. You've forgotten how to lobby. You don't have a prestige product. You have no clue how to defend yourself, given that the Trumpies don't really speak our language. I'm here to help, and I'm one of you, and you've forgotten this. There is an entire meme that does-- makes no sense, that I am somehow a grifter, charlatan, troll. I don't even know what these words really mean. It comes out of five people, and it has rotted your brains, and it is part of the reason why it was controversial. I couldn't speak in the physics department. I had to speak in the Hart Seminar. In the physics department, because Lord knows the physics department of the great University of Texas at Austin shouldn't be sullied by my presence. This is absolutely the most offensive thing you can do to a speaker.

00:36:37,240 --> 00:36:37,900 [speaker_5]

[laughs]

00:36:37,940 --> 00:39:00,460 [speaker_0]

And I'll say what I said to Sean Carroll, okay? And it, it, it got a lot of meme play. How dare you? How dare you is a sentence inside of a communal structure with no one. You have lost your norms. That's the problem. In order to restore ourselves, we have to realize collegiality and decency and money is a weapon that we will need to keep the world free and prosperous. Essentially, when they ask me, "Why should our taxpayer dollars be used to fund theoretical physics?" I turn the question around. Those are our physics dollars. You have the best licensing deal in history. We gave you, what? The semiconductor, Wi-Fi, electron orbitals for chemistry. What haven't we done for you? We've kept the world safe with the most powerful weapons that brought peace to Western Europe that never-- has never been seen. We are dangerous. You should be paying attention to us, and you're ignoring us and trifling with us. Those are our dollars. You have the best deal. Shut up. Give us what we need, and we will continue to make you rich and powerful. But to not cut us in on the wealth and prosperity that we have provided to you is an abomination. I'm gonna say something. You produce something that has a technical name called a public good. A public good is something that is inexhaustible and inexcludable. If I have a lighthouse, I can't only charge the ships that pay into the lighthouse to receive its beacon. If I had a digital file, it would have an infinite number of plays. It doesn't wear out like a vinyl record. Anything that has those twin attributes, even a free market economist who's any good or, or fair will know that price and value can't be held together in a market. You produce a public good. That is your claim on the taxpayer, is that the market can't see what we do. And you don't know how to defend yourself because you don't have the language. So what I'm trying to do is to say, I am here as an angry best friend.

00:39:00,710 --> 00:39:01,340 [speaker_5]

[laughs]

00:39:01,400 --> 00:39:14,500 [speaker_0]

I'm pissed as hell at how you conduct yourselves poorly. You do the most important work, and you don't realize how much danger you, and I would say we, but you wanna exclude me, then it's just you who are in danger 'cause I'm gonna be fine.

00:39:14,520 --> 00:39:14,880 [speaker_5]

[laughs]

00:39:14,980 --> 00:39:38,120 [speaker_0]

I want to help. I am not Sabina. Sabina wants to cut off your funding. My feeling is if the worse a person is to me, the more I probably want to jack their funding up because a lot of that is what happens when we are operating below the amount of nutrient that we need in the system. We start to see each other as a source of fruit. The last thing I'm gonna say before the discussion is

I was just in the urinal-

00:39:38,720 --> 00:39:38,729 [speaker_5]

[laughs]

00:39:38,729 --> 00:39:58,760 [speaker_0]

And I saw it says, "Stop Trump." That was not interesting. Right under stop Trump, barely visible, it says, "Elaborate," with an arrow [audience laughing] on stop Trump. Doesn't say you shouldn't probably stop Trump. It just says, "I need a little bit more."

00:39:59,520 --> 00:40:00,820 [speaker_5]

[laughs]

00:40:00,880 --> 00:40:55,240 [speaker_0]

So what I'm going to say is I don't care whether you're for Trump or against Trump. We have to survive through any administration. Think about the office. If you can't honor the participant in the office, think about the office. This is happening in America. It's done with American pa-taxpayer dollars. This is an American university. It's somewhat uncomfortable because there's a way in which science is international, but there's also a way in which we are the most important community inside of a university. Uh, physics first, biology, mathematics after that, in my opinion. And we have to lead when nobody's doing it. So the hope is to start a discussion about that. Thank you for your time. [audience clapping] Andreas, can I ask you to help moderate?

00:40:55,680 --> 00:41:02,010 [speaker_3]

I was about to get up to do it just then. Um, do we have questions? I hope we do. Uh, let's start here and-

00:41:02,780 --> 00:41:21,880 [speaker_6]

Yes. Um, do I understand correctly that your assertion is that the influx of foreign-born physicists in the US after nineteen ninety is due to the result of an attempt to depress scientific wages in the United States, and that your intended remedy is to reduce the overall number of scientific workers in the United States?

00:41:25,120 --> 00:44:35,348 [speaker_0]

My goal is to engineer our markets so that our foreign friends and colleagues and workers are not usable as a wage depression mechanism. In other words, I have a right. That right is called the right of first refusal. There's a job that's offered. It is traditional that Americans have first access to their own labor market. Any right like that is-- That, that right is the basis of your wealth. So you do one job here, you do the same job in Paraguay. Why is it so much more valuable here? Because you have access to a labor market that in some sense keeps the wealth. This is our distribution mechanism to make sure that everybody gets richer. What happens when you call that red tape? Did you offer this job to an American? Is there no qualified American? Okay, well, it's a right. The work of Ronald Coase says that when a right is made saleable, the market comes to the same equilibrium. So for example, let us imagine that you are a polluting factory and I'm a fisherman. If we make the right to pollute the lake a right, and we give it to the factory, the fishermen will buy the right because they don't want as much pollution in the lake. If we give the right to the fishermen, the factory will buy the right because they wanna be able to produce something, and a factory can't work producing no pollution. The key point of Coase, one of the most brilliant thing, maybe the most brilliant thing in the history of economics, is he said it doesn't matter which-- who you give the right to. The resource will be used the same way, but it does matter who gets rich. So my feeling is, if you want to import all these best and brightest people, Americans deserve to sell that right to their employers. So I wrote a paper in two thousand and two, um, called Migration for the Benefit of All. Those of you

who spend way too much time on the internet may know that I am the author of the Great Replacement Theory, a white nationalist narrative. And I was instrumental in ratifying the world to dilute the white race. Um, very strange how people come to completely insane conclusions. Rights should not be taken away from people uncompensated. And my claim is, if markets are what keep us free, let us participate in those markets. I would love to visit you in your second home with your three children through college with a full retirement. And I will point out that the professors that I met at the beginning of my career were at the end of theirs in beautiful homes in places like the Vineyard. Um, duchess, if you will. We have had a tremendous fall from the times that I used to swim, um, sun worshipping with Ralph Lauren, uh, off of, uh, off of the Vineyard.

00:44:35,828 --> 00:44:36,048 [speaker_6]

S-sorry.

00:44:36,068 --> 00:44:37,748 [speaker_0]

We need to get back to being rich.

00:44:38,028 --> 00:44:42,078 [speaker_6]

Can, can you narrow that into a more... I'm having a hard time picturing what that policy is.

00:44:42,078 --> 00:44:50,868 [speaker_0]

Read my paper from two thousand and two on how to make rights saleable and so that they're not taken from us and used as wage depression mechanisms.

00:44:51,268 --> 00:44:51,388 [speaker_6]

Hmm.

00:44:51,568 --> 00:44:52,588 [speaker_0]

Happy to talk about that.

00:44:54,208 --> 00:44:54,788 [speaker_7]

Over here.

00:44:55,228 --> 00:44:55,428 [speaker_0]

Yeah.

00:44:56,488 --> 00:45:41,068 [speaker_7]

Yes. I feel during your talk you made a bit of a conflation between, you know, theoretical, you know, high energy physics, if you call it, with the field of physics as a whole. Physics is an extremely broad field, uh, and the, the gifts that you say that physics has brought us, the transistor, um, the, you know, the electron orbitals and such, you know, those don't-- didn't come from experimental high energy physics. They came from the other-- Excuse me, the theoretical high energy physics. They came from the other aspects of physics. So, you know, how do you think your message here changes if you, you know, take, take the field of physics as a whole, uh, or do you include kind of the experimental result, you know, experimental side of physics more so than the theoretical side of physics?

00:45:41,248 --> 00:46:58,518 [speaker_0]

Great question. Well, quite true. And there are parallels in other fields. For example, evolutionary theory is very close to public good, and molecular biology can often be done as biotech in Memphis Square right outside of MIT. My claim is that most of the experimental stuff is a lot closer to industry. It's a lot closer to being concrete. You guys are not nearly in the desperate situation that your theoretical friends are in. And my feeling about it, before you give

my talk to the experimentalists, there's a, there's a project that I sort of have with Brian Keating, which is why is it that all the podcast hosts wanna talk to theorists? And if you recall, I, I brought Duncan Trussell and Kurt Metzger, and at the Comedy Mothership in the green room last night, everybody was waxing rhapsodic about the experimentalists and their crazy toys and how, you know, seventy-seven degrees above absolute zero and, uh, humming millibars and all this kind of stuff. So I've got your back. All I'm trying to say is you're not in the same desperate situation because what you're doing-

00:47:00,448 --> 00:47:08,528 [speaker_7]

You say we're not in a desperate situation, but my advisor says we can't run experiments at liquid helium temperature because we don't have enough money to buy helium. So that-

00:47:08,578 --> 00:47:08,628 [speaker_6]

Yeah

00:47:08,848 --> 00:47:13,177 [speaker_7]

... we're, we're desperate. It costs a lot of money to run a lab, and we don't have, like, we don't have money.

00:47:13,177 --> 00:47:17,988 [speaker_0]

Four words: How can I help? I'm gonna be here after the talk.

00:47:18,028 --> 00:47:20,618 [speaker_7]

You got your checkbook? [laughing]

00:47:22,378 --> 00:47:28,068 [speaker_0]

I'm sorry. What did you think of that question? Hmm? You just asked a question.

00:47:28,908 --> 00:47:32,088 [speaker_7]

You got your checkbook? [laughing]

00:47:32,288 --> 00:47:37,128 [speaker_0]

It's an obnoxious question because I know how you see me.

00:47:38,628 --> 00:47:38,788 [speaker_7]

No.

00:47:39,588 --> 00:47:48,918 [speaker_0]

That's a nervous question. It's a nervous question because you don't know how many zeros I have to my name. I was Peter Thiel's managing director, okay? Yeah.

00:47:50,308 --> 00:47:53,858 [speaker_7]

I don't mean you personally, right? What would help is if we got, you know-

00:47:53,858 --> 00:47:54,408 [speaker_0]

But you're not hearing me.

00:47:54,748 --> 00:47:55,428 [speaker_7]

What?

00:47:55,468 --> 00:49:11,364 [speaker_0]

You're not hearing me. They see you coming a mile away when you pull that stuff. They see you as not actually being taken seriously as a mind. They see you're not treating me like a Harvard

PhD. You're treating me as a rich guy. How about writing a check, Eric? No, I'm not gonna write you a check. I wanna get you billions, and you're pestering me for what? Don't do that. You're giving away your power. You're giving away your power because other people see through what you just did. When we say things like pitch me, bro, the point is, what are you gonna do for me? What do I get out of this? You gonna name a particle after me? I don't even know what you do, but you know, my, my question is, it is not my job to fund you individually. You have the wrong incidence. You ever look at a high heel that comes to a point? The amount of pressure that's on that point is wrong. You look at something like an espadrille, and it's spread out. Okay? I am not your funding source. You know more about that.

00:49:11,404 --> 00:49:11,774 [speaker_8]

[laughing]

00:49:11,844 --> 00:49:12,864 [speaker_0]

You have a question.

00:49:12,904 --> 00:49:25,264 [speaker_9]

Hi, Eric. Uh, you talk about a lack of representation and the idea that the field has, uh, insufficient representatives to get the right money, the right representation, and value for their work.

00:49:25,704 --> 00:49:26,984 [speaker_0]

Let, let's start with that, tourism.

00:49:28,724 --> 00:49:34,634 [speaker_9]

Yeah. Who do you think are the best representatives that exist today? That list that you gave, you had criticisms of.

00:49:34,874 --> 00:49:34,904 [speaker_0]

Yeah.

00:49:34,934 --> 00:49:37,424 [speaker_9]

But who are the people that can potentially do it right?

00:49:39,184 --> 00:52:36,324 [speaker_0]

I'm gonna be horrible about it. I think Nima is a special person. Nima has all the energy, all the enthusiasm. He's a genius. He's really, really smart. He's got a very broad love of stuff. He's very adventurous. I can talk to him about Casimir effects and geckos and potentially going faster than the speed of light. And he's also really circumspect. So he's like, in terms of psychological characteristics, he's high in trait openness, and he's highly disagreeable. Perfect two traits. I'm high in openness, high in disagreeability. So he's my man. Um, Sean Carroll would've been great. What is that guy's problem? I don't know. This is not personal. I have no desire to destroy Sean Carroll. He didn't come after me, I wouldn't go after him. You wanna know who else? I didn't put up me with Tulsì Gabbard, me with RFK Jr., me with J.D. Vance, me with the incoming head of NSF, uh, Jim O'Neill, me with Jay Bhattacharya at NIH. I am willing, tentatively, to broker this, and I don't have any ego in me, and I don't benefit from it, if you would stop drilling holes in my boat. I cannot defend you when you are at your worst. So my feeling is I would put myself generously. There's one reason I don't want to record in here. I am offering to broker this if I can find representatives of the community who are willing to do a, uh, the McAfee on... You need a Symantec on your malware. There is malware. It just is. You can't see it. You're part of the cult. Assume that you can get rid of enough malware, because nobody can get rid of it all. I could name you the people who I think are fantastic. Why aren't we using David Tong, even if he's over across the pond? Fantastic. Frank Wilczek is old, but he's good.

He's solid. And if you want to be incredibly horrible about it, the problem really stems from six or seven individuals. You know who they are. It's the only game in town crowd who... An old story. I, I'm in the Harvard Physics Department library. It's beautiful. It's a top floor. And I'm asking questions to somebody, and he says, "Oh, this is a strange question." I said, "Well, I'm a, I'm a mathematician." He said, "Oh, a mathematician." This was way back in the day. He said, "We found out two things by talking to mathematicians. Only two." And I said, "What are they?" He said, "One, that we are much, much, much, much smarter than everybody we talk about." I said, "Eh, and number two, dare I ask?" He said, "Our notation sucks."

00:52:37,164 --> 00:52:37,233 [speaker_8]
[laughing]

00:52:37,233 --> 00:54:13,344 [speaker_0]
Now, those sorts of interactions, they're quaint when they're individual and quirky. But the Strominger-Green-Witten, Jeff Harvey, Michael Duff, Leonard Susskind, strings, strings, strings thing was a total flip. And what's important is to take the people who can think properly and who have some standing and have the ability to speak without too much um and too much ah. You know, I'd love to get-- I, I'd, I'd love to get Michael Peskin involved. But there are all sorts of people I can point you to who we should be convening to represent the field. And by the way, I want mathematicians. I can tell you who I would want in mathematics. I'm a huge Ed Frenkel fan. The Soviets are a great resource because they hate communism. They see through the bullshit in the US because they had to live through it in the Soviet Union, and they've kept the fire, and they're high prestige, and they're high energy. I like drinking with them. No offense. Anyways, anyways, my point is we have lots of people to represent us. And to be entirely honest, um, Jacques has been doing an amazing job with that blog Musings, which has slowed down in recent years. But he's one of the OGs of science communication. I think that, you know, I, I'm a huge Kathy Freese fan. Uh, Scott Aaronson in this department. Just local talent, right? You've got a lot of firepower. You guys at the Hollywood Wine Bureau can do it. Active. Oh, and Sudarshan.

00:54:14,364 --> 00:54:16,544 [speaker_9]
But what's preventing that coordination today?

00:54:17,724 --> 00:54:45,732 [speaker_0]
You don't know that you're in danger. You don't know that you can fight it. You don't know who to call. You don't have people's personal phone numbers and direct lines. You, you're, you're passively watching your world dismantle as the boiling frog analogy. You've lost so much since something called the Mansfield-- How many of you know what the Mansfield Act is? Do you know it through me? All right, I owe you a beer.

00:54:46,632 --> 00:54:47,372 [speaker_1]
The Mansfield what?

00:54:47,812 --> 00:54:48,592 [speaker_0]
Mansfield Amendment.

00:54:49,092 --> 00:54:52,362 [speaker_1]
Oh, yes. Uh, yes, the one named after Mike Mansfield of Montana.

00:54:54,972 --> 00:58:55,162 [speaker_0]
What it said was-- What happened, I believe, was there was a walkout in MIT physics department, where Manhattan Project-era physicists, angry at being associated with the government during the Vietnam War, staged a walkout in May of nineteen sixty-nine. And the

main recipient, I think, of federal funding was something called the Instrument Lab at MIT. Somebody-- Anybody can correct me if you have an AI. And immediately, the federal government got really pissed off, as it recently did with Anthropic having too much morality to be a contractor. And so what happened was that MIT moved the Instrument Lab out and became Draper Laboratories. You have Draper Laboratories, Lincoln Laboratories, where all the defense kind of stuff is done. And then promptly, Mike Mansfield, um, authored an amendment which said that you couldn't fund blue-sky research out of the Defense Department unless there was express purpose that was military to stop the military having good relations. So the way things are supposed to work, just to be clear, is that we had genius-level administrators back in the forties during the days of the bomb and the war. And they set up a cryptic system that nobody quite explained to the public. For example, overhead isn't overhead. You guys know about the direct costs and how overhead works? All too well. What? All too well. Okay. [laughs] You apply for a grant. The question is, how do you stop-- Let's create a university that I'm sure doesn't exist. Um, East Virginia-- The University of East Virginia. Uh, the University of East Virginia is a terrible school. The federal government wants to give money to Princeton and the University of Chicago and the University of Texas at Austin, and it doesn't want to give money to the University of East Virginia. So what it says is we'll have a competitive grant process, and we'll give a giant chunk if you hire people who can get grants, and we'll give it to you as overhead. And that way, the federal government will keep Harvard, which is rich as hell, as a federally funded university that will be ostensibly private. So it's completely cryptic. It doesn't make any sense. This is a little bit like, um, looking through USAID and seeing all these bizarre line items, right? So, you know, if you- Mm-hmm ... if you decide that you're gonna spend, uh, three million dollars on prostitute education in Bulgaria, if you read that, you can't tell what that is. But what it is is an attempt to destabilize Bulgaria using soft power of nonprofits. So all of these ways- Got it ... in which the government actually works as opposed to how it ostensibly works are confusing. Now, what we're doing is we're seeing crazy levels of overhead. That isn't crazy. That's supposed to be the federal government supporting our richest universities as, um, custodians of the American science and engineering enterprise. And what the-- I hate to say this. We, the American physics community, never use this word, are outgrowths of the defense industry and the Defense Department. And partially, that's good because one of the things is that when they misbehave, we are supposed to be able to exert power on them. We're supposed to be able to do course correction. And we don't do this in part because we're too pure. So we have a very delicate balance where we have to recognize that we have a secret university system called the National Labs. We have secret labs that are hung off of regular universities, like the Applied Physics Group at Johns Hopkins. Defense is all through what we do, even though we have the purest and noblest of motives. And everybody needs to be talking about this because this is who we are. If you want to avoid it, feel free to talk about it. We should talk about it. That's more precise. So I'll take this hand up.

00:58:55,912 --> 00:59:34,612 [speaker_10]

Yeah. I'm kind of a more pure guy. Um, at the beginning of your talk, you had said a few comments about how you actually are a fan of some form of gatekeeping. It, it seems like during this talk, you kind of wanted us to see how a lot of the notions we have around how institutions historically work are kind of these, these fictions. I'd be kind of curious to know just sort of broadly what you think like scientific research institutions should aim at, how they should work, and maybe more particularly, how should we manage dialogue in those communities? How should we correctly sort of go about verifying what we consider to be correct, and how do we do gatekeeping in a way that's institutionally healthy? Sorry if that's-

00:59:35,172 --> 01:01:20,259 [speaker_0]

No, it's a fair question. Let me, let me try and answer it. And again, most of us are not used to difficult questions in university. This is a lot about taste. It's a lot about decency. It's about adultery. It's about recognizing-- You heard what I had to say about Sean Carroll. You can

imagine how livid I am that you hear me praise him. Jacques has said some not very nice things about him. Andreas as well. [laughs] As long as they're fair, I'm supposed to have a thick skin, and I have to thank my colleagues for course correction. When they're unfair, as what Sean Carroll did to me, I'm, I'm, I'm gonna curb-stomp you. But what I'll do first is offer you a hand up because I don't really want permanent-- I-- We don't get anything out of feuding. There's a marvelous story about Feynman in the Manhattan Project where he looks at this bitter argument between all these great scientists, and he says, at the end, everybody agreed he was right. This is a muscle. It's supposed to be taught. It's supposed-- You're supposed to learn this with your mother's milk. Being a scientist is so much more than the ability to solve equations. Being a scientist is a deep personal commitment to truth at a level that is painful for almost anyone, and it's supposed to come from your elders. You're supposed to get slapped and spanked when you're trying to feel your oats, when you're prancing around like, you know, you're the shit. We don't have professors who know how to do that anymore. Everybody knows what it feels like to be mentored by something and someone amazing. Who, who's seen Whiplash?

01:01:22,340 --> 01:01:23,110 [speaker_3]

Yeah, I, I-

01:01:23,110 --> 01:01:32,740 [speaker_0]

I have not. I've seen all the others. [laughing] That guy is abusive. But the question is, is he the right kind of abusive? Are you-

01:01:32,900 --> 01:01:33,000 [speaker_3]

No

01:01:33,040 --> 01:01:37,360 [speaker_0]

... doing this-- do you understand that this is the hardest thing you can do with the human mind?

01:01:37,549 --> 01:01:39,000 [speaker_3]

I- [laughs]

01:01:39,020 --> 01:04:24,420 [speaker_0]

Do you understand that this is the most demanding thing in the world? You know, there's an old saying that, uh, if you can do anything other than write, do so. But a writer is somebody who can't do anything other than write. It, it has to be a whole life commitment to wanna do this thing because it's so hard. And part of what makes it hard is saying, "I was wrong." What you see in the string community and the quantum gravity community is a congenital inability for the leaders to admit that they're wrong. And I don't know what to do about that because you had this with Lysenkoism in the Soviet Union. It fell to the physicists under Stalin to get the biologists back on track after they'd had their minds hollowed out by sort of a, a form of the Marxism. So it, it's supposed to be taught to you, and you're supposed to have an advisor. That advisor is supposed to be your doctor father. It's supposed to be a father figure or a mo-mother figure who teaches you how to be a colleague. And I'm gonna say one more thing. You need money. If you don't have enough money, people have to die. People who are trained, who are good, have to be exited. You need enough money to support the people who deserve to be here, who invest. So this is an investment quality education. And without that money, I can teach you the ethics, but it's going to end up as each of us eyeing each other as a source of protein. When I was in Berchida years ago at a physics conference, an old guy said-- he was hearing us all nervous about our job prospects. He says, "Let me just..." He sat down. He said, "You're all worried about whether you're gonna get employment. In my day, we didn't apply for jobs." He said, "We didn't even apply for jobs. It was as simple as that. We didn't apply for jobs."

If you were any good, a professor got on the phone, picked up the phone, called somebody, and you got a job at the end of the call." Bring back slush funds. Bring back a lack of accountability. Stop with the transparency. I know that you've been told all these things are good and necessary and whatever. These are internal matters for our community because nobody else understands what we do. People think that Sean Carroll humiliated me because they can't follow what happened. They listen for the smugness and they look at the facial expressions. We are the only ones who can adjudicate this. And it's just really important to recognize that we need autonomy, we need money, and we need collegiality, decency, and professional ethics are a weapon. A community with professional ethics will out-compete one that does not have them.

01:04:25,180 --> 01:04:29,120 [speaker_3]

What about my doctor father thinks? Uh, he's an outside father, so [laughs]

01:04:29,200 --> 01:05:41,300 [speaker_11]

So I wanted to, uh, get back to, uh, the, the topic of faculty immigration. Um, a-and I, I thought I would start with a story, uh, which when I was about the age of many of the graduate students here, a young man trying to think about what I wanted to do and where I wanted to do it, the Canadian Minister of State for Science Tech-- and Technology at the time gave a radio interview. And what he said was, in response to a question: "Basic research? We don't need to do that. We can let the Americans fund it." At which point I sort of said to myself, "Well, I guess I know where I'm going." Um, fast-- A-and, you know, I was on every possible visa, F-1, J-1, H-1B, green card for thirty-three years till I finally became a US citizen a year ago. Um, and nowadays, I, I think, you know, for all these guys here, um, uh, uh, when you're trying to figure out what you're gonna do and where you're gonna do it, um, my advice would be go to Canada-

01:05:41,840 --> 01:05:41,970 [speaker_3]

[laughs]

01:05:42,070 --> 01:05:48,120 [speaker_11]

... or Europe, not the UK, Europe or Japan, or pretty much anyplace else.

01:05:48,500 --> 01:05:48,920 [speaker_0]

Or India.

01:05:49,480 --> 01:05:52,530 [speaker_11]

Or India even, yeah. Or, uh, I mean, or if you can get c-

01:05:52,530 --> 01:05:56,780 [speaker_0]

I'm getting my overseas citizen of India certification as best as I can.

01:05:57,120 --> 01:06:11,540 [speaker_11]

So, so I, I mean, I think, um, that, uh, this-- that the, the, the influx of foreigners taking jobs in physics at US institutions-

01:06:11,920 --> 01:06:11,929 [speaker_3]

[laughs]

01:06:11,940 --> 01:06:21,320 [speaker_11]

... is not necessarily a bad thing because what it means a-a-at, at, at zeroth order is that this is the best place in the world to be.

01:06:21,550 --> 01:06:21,550 [speaker_0]

Yeah.

01:06:21,580 --> 01:06:39,820 [speaker_11]

And they all wanna be here. And I think that as, if you're honest with yourselves, has ceased to be true. Um, and, um, one of my students, uh, sitting somewhere in this room who, uh, is grad-- getting his PhD this year, uh, he is going to the UK for a postdoc.

01:06:39,900 --> 01:06:41,780 [speaker_0]

China is very active in trying to lure-

01:06:41,980 --> 01:06:42,400 [speaker_11]

Oh, I know

01:06:42,420 --> 01:06:43,100 [speaker_0]

... its people.

01:06:43,160 --> 01:06:43,420 [speaker_11]

Yeah.

01:06:43,700 --> 01:06:43,800 [speaker_0]

So-

01:06:44,060 --> 01:06:54,380 [speaker_11]

So, a-and anyways, I, I just wanna bring that as, as, as sort of a counterpoint, uh, uh, to, you know, it's not just about the wages, and it's not, uh, uh, th-that's

01:06:55,372 --> 01:07:25,532 [speaker_0]

You know, a, a, a higher order effect in, in at least to my way of thinking is to like where you want-- where are you gonna be in twenty years? We have to talk about the economics of labor markets and the political economy of science. And I believe very strongly that... You know, look, I-- To keep my Hindi as worthy of, I go into lots of rooms on Twitter Spaces.

01:07:25,952 --> 01:07:26,012 [speaker_12]

Yeah.

01:07:26,052 --> 01:07:30,212 [speaker_0]

And occasionally I get called up, you know. So I've got all these Indians talking about H-1Bs.

01:07:30,652 --> 01:07:31,432 [speaker_12]

Yeah.

01:07:31,532 --> 01:09:08,392 [speaker_0]

I'm a huge critic. I hate the H-1B terribly. People are like, "I'll have your nose, Mike." It has nothing to do with your friend, the foreigner, or my wife, the immigrant, whoever. We need labor markets, and we need structures so that we don't get destroyed. Right? The key point is this has been engineered to use our foreign colleagues to depress our bargaining position at the table. It was engineered by Miles Boyle, who's still alive. Go find him. What I'm trying to tell you is this isn't a question of do you like foreigners? Do you not like foreigners? Is science international? I have a feeling he wasn't born in America. [laughing] You've got to, you've got to, you've got to decouple those conversations. The conversation about the structuring of our immigration programs and our immigrant colleagues are not the same conversation. You know, I, I in fact coined the term the Sudarshan effect for the unfair way that we treat people of extraordinary ability who are brown and cute. We just have to be very careful and learn how to do this with decency and respect, and that's it. And by the way, I will point out that when I want to go to a seminar, uh, at TIFR, I have to go through a military checkpoint. You know? It's like

they understand. I understand. You know, it brought us Einstein and Klaus Fuchs. It's a double-edged sword. We just have to be grown up about it. That's it. Not-- It's not personal. Another question.

01:09:09,152 --> 01:09:24,692 [speaker_12]

Um, [clears throat] so for better or for worse, computers are getting very good at math, and they're probably gonna come for theoretical physics next. And so, I mean, how can we really justify how much we're being paid when they can, you know, essentially do our work that we can for a fraction of the price?

01:09:25,812 --> 01:09:26,412 [speaker_0]

Great question.

01:09:27,032 --> 01:09:27,792 [speaker_12]

Yeah.

01:09:29,972 --> 01:14:05,404 [speaker_0]

Because I am so idiosyncratic... Look, geometric unity is one hundred percent one person. That means when I query these models about geometric unity, and they know different amounts about geometric unity, I know that I am being rigged as a puppet inside your machine. So if you ask about geometric unity to Claude, and then you go over to Grok and all these things, that thing is taking me and animating me. Right? By the way, I actually heard, I don't know if you guys saw this, in this fake-- a trailer for a fake movie that doesn't exist with Neil deGrasse Tyson. Who just saw Neil deGrasse Tyson in front of me in a diner in the rain? [laughing] "Neil, it's not what you think." You know? Um, this is taking us and operating it for free. First thing is I wanna get paid, and I don't wanna get paid individually. I want my community to get paid because we all share in a lot of this. It happens that I can isolate my own voice. By the way, there are entire podcasts with me where I did not agree to the podcast, and I never appeared. My wife says to me, "Eric, why are you talking to Lex Fridman as if you're a commercial fisherman?" And I said, "That's not really me." Um, we've got to start going after Sam Altman and Elon and all of these people. Now, Sam is a friend of mine. And when I-- Sam once said to me, "Eric, you're not paying enough attention to AI. It's a lot closer than you thought." And I really changed my mind because of Sam. So then I called him up at some point and said, "I believe the unified theory is a lot closer than you think." And he immediately flew down to Ben Gurion Airport and spent an hour and a half with me. And one of the things he said to me, some things I can't repeat, is what you do and what I do both should be done on a military base, but we're just not configured for that very long. What we do is very dangerous. We're not getting paid for it. It's being stolen from us. And that has nothing to do with how we need to adapt because this is a tool. And right now it's actually not very good. The tool is not very good. So if you wanted to... I even think it's possible that it's gonna hit an asymptote and not get to the highest end stuff that we give it. On this round, so the middle-- at the moment, my view from Silicon Valley is you're in a hype bubble, and this hype bubble is about scaling one insight, which is that all meaning can be converted into small angles in large vector space. You take anything in the world, you tokenize it, which is this layer that we don't even really talk much about. You put it in a relatively large vector space. Generically, any two vectors are all equivalent. They could almost certainly be equivalent. If two things are meaningful to each other in some inner product, we say one attends to the other. You know, there's attention mechanisms, and those are just basically two tensors. And the idea is that that turned out to be enough to mimic almost all of what it is that we do as intelligent beings, because most of the time we're just LLMs. Occasionally, we do something interesting, but ninety-nine percent of your day is just an LLM. This conversation is mostly LLMs talking. So it was able to pass the Turing test because of that. This hype cycle is about scaling that one dumb insight. When that runs out, the only reason that these valuations are as high as they are is that Wall Street is salivating about the

ability to replace all of you. That is the dream. I've shared with you universal high energy. The whole idea is finally we can get rid of our population and capital can be unshackled from labor. Defend yourself. Learn posts. Learn posts, posts, posts, not UBI, posts. Markets are freedom. Get markets to work for you. You're about to be obliterated. Learn how to defend yourself. When somebody else says red tape, you just say, "Red tape? That's my livelihood. It's a right. You wanna take it away because you're claiming that it's an impediment to free markets? No, no, no. You buy it from me. Let's talk about a price." You guys don't know how to defend yourselves, and that's part of the problem. Imagine if you collectively hired, I don't know, Sullivan & Cromwell or the best ad agency or whatever, and you learned how to play those dirty games through your representatives so that you're not sullied. You don't know how to do that. I would love to help, but in part, you better, you better fight what's gonna be taken.

01:14:06,684 --> 01:14:07,224 [speaker_13]

You have time.

01:14:08,364 --> 01:15:33,794 [speaker_14]

So I feel that there is some misunderstanding about, like, our experience as scientists because my colleagues at National Labs who have been there from the mid-century, they were promised they just have a salary, and they have to just produce good work. But the experience now for scientists is that we are trying so hard to keep our heads above water, and every time there is an administration change and a policy change and budget changes, our grants get cut suddenly, and we are having to shift, and we tally all of our work by the hour per particular grant and project. And I feel like you are dangling this idea that you could help us because so much of our job is like, how do I produce a paper while also writing a grant, while also doing my travel and doing all the bureaucratic work around it and talking to my other colleagues at academic institutions and elsewhere and m- mentoring my students and teaching? And clearly, you care about this because you wrote that whole paper. Um, so why don't you just help? Is this transactional? Do you just want the acceptance of physicists so that then you would help us? The burden of proof is on you to create a document that sufficiently s- satisfies everyone that you have proven something new, and I don't see that from the scientific community, and-

01:15:34,184 --> 01:15:36,384 [speaker_0]

Sorry. I lost what you said.

01:15:36,424 --> 01:15:41,964 [speaker_14]

What I'm-- The burden of proof falls on the person who is trying to suggest something new to the community.

01:15:42,124 --> 01:15:42,524 [speaker_0]

So it's yours.

01:15:44,404 --> 01:15:44,564 [speaker_14]

If-

01:15:45,484 --> 01:15:46,904 [speaker_0]

You mean GDU, right? Okay.

01:15:47,344 --> 01:15:51,544 [speaker_14]

I just mean w- w- what your paper that you have put out, that you were showing us a clip about.

01:15:51,684 --> 01:15:51,964 [speaker_0]

No.

01:15:53,164 --> 01:16:02,484 [speaker_14]

If you put out a paper, if I put out a paper, the burden of proof is on me to show what I claim I am showing and to convince my colleagues. I'm not seeing this-

01:16:02,504 --> 01:16:03,284 [speaker_0]

Am I your colleague?

01:16:03,984 --> 01:16:06,524 [speaker_14]

Well, you s- keep using us and we and our.

01:16:06,604 --> 01:16:26,704 [speaker_0]

I don't do that normally. We normally talk... I, I am breaking all of my rules. I, I, I was a visitor for a week at the Perimeter Institute. I was not supposed to say that I wasn't at the Perimeter. I gave a talk to the University of Chicago. They have AV equipment, which is how you film it. It's like I'm your side piece.

01:16:27,064 --> 01:16:28,314 [speaker_14]

[laughs] Okay, we-

01:16:28,314 --> 01:16:30,904 [speaker_0]

I'm some hot chick you can't take to the Christmas party.

01:16:31,084 --> 01:16:31,444 [speaker_13]

Um.

01:16:31,544 --> 01:18:05,264 [speaker_0]

What I'm trying to say is it's not personal. I don't wanna confront your spouse. I'm saying get over yourselves because this is not a sellable product as is. I am already pre- plenty prepared to take a massive advance on your future success and massive cost on my own shoulders. My mom once told me that she is a lifeguard. They're taught not to save a drowning person unless the drowning person agrees to cooperate with the rescue. What you are doing is so catastrophically stupid. And I'm saying this as, as a loving friend. You-- It is so embittering, so completely unnecessary, so divisive, so invidious that you have not calculated, just as the gentleman over there did, what the effect is of saying, "You're only here 'cause you're rich and famous. The burden of proof is on you." The community is not functional. I'm not just talking about getting you funded. I'm also try- talking about getting you back into fighting shape. I'm talking about going to the gym. I'm talking about rehab. I'm talking about physical therapy. I'm talking about pain. And my feeling about this is if you're interested, I'm willing to advance a tremendous amount on your behalf. If what you want is to study ADS-CFT and amplitude heavens and have Meecher, Cockcroft, Johnon write God knows what that nobody cares about, I have no interest. I just don't. There has to be a recognition of failure because that is part of science.

01:18:05,944 --> 01:18:16,584 [speaker_14]

But you're saying that the best work that our community has done was when scientists were able to just freely study their science without having to appeal to anybody.

01:18:16,824 --> 01:19:43,552 [speaker_0]

That's right. And do you know how you got there? This was a tiny boutique community, and you destroyed two cities in Japan. It got everyone's attention, right? And then you came out with lasers and semiconductors and the World Wide Web. Like, you could do anything. We just believed wholeheartedly in your genius and your brilliance, and we gave you money, and you were part of a great arrangement. We were part of a great arrangement. Let me say it as this,

okay? I, I, I am, I am the black sheep of the family come home. They believed in our ability to think out of anything, including anti-gravity, right? And the claim is, is that nobody's done anything so insane like tomography. You know, the big of-- the big of events have been machine learning. Machine learning and distributed computing through blockchain. These are these giant advances. I want things to come out of this community that brings everyone to their knees so that we never have to advertise just how badass we actually are, and we aren't doing those things.

01:19:43,652 --> 01:19:49,552 [speaker_14]

I'm not convinced you care about us. I'm convinced that you want our acceptance, and once you have that, you will help us.

01:19:50,252 --> 01:20:09,122 [speaker_0]

Why don't you form a group, hold that belief and explore it? And then other people who actually believe that I am sacrificing my credibility in my own community, and my credibility in your community, and my credibility in the outside world. Like, why don't you look at revealed preference and figure out what I actually care about? I'm happy to talk to you about that.

01:20:09,172 --> 01:20:12,672 [speaker_14]

What, what are you risking? How are you risking it by being here?

01:20:14,892 --> 01:20:20,292 [speaker_0]

Have you ever looked at what is said about me on these completely lying memes?

01:20:20,312 --> 01:20:20,912 [speaker_14]

[laughs]

01:20:20,952 --> 01:20:27,472 [speaker_0]

Do you, do you have any idea how painful it is to wake up with your parents saying, "Oh my God, Eric, you did what?"

01:20:27,832 --> 01:20:28,572 [speaker_14]

[laughs]

01:20:29,632 --> 01:20:54,402 [speaker_0]

I, I-- Literally, I'm, I'm so many people's goblin. It's a, it's a lousy, thankless life. And, and what are you doing? You're adding to it. And my claim is, is that there's a muscle in your head where you don't realize I'm a visitor who obviously cares about your community, and you're embittering me. Why? I'm, I'm just curious. What's your motivation? What do you think you're doing?

01:20:55,052 --> 01:21:01,192 [speaker_14]

I think I want to understand if you think that physicists, especially experimental physicists, for example-

01:21:01,432 --> 01:21:01,542 [speaker_0]

Yeah

01:21:01,712 --> 01:21:08,792 [speaker_14]

... who do work closely with theoretical physicists, why wouldn't you suggest them to the scientific advisory committee if you-- If you feel that there's a good-

01:21:08,912 --> 01:21:10,482 [speaker_0]

I am trying to get-- I'm tr-

01:21:10,732 --> 01:21:16,751 [speaker_14]

You said that you didn't have anyone to suggest that, that was a physicist to the committee when called upon.

01:21:17,531 --> 01:21:40,702 [speaker_0]

Look, I wanna get Brian Keating on these committees. All right? Brian Keating is an experimentalist. He's a great friend of mine. We fight about all sorts of stuff. Um, I brought, I brought you your podcast. I'm not gonna justify myself to you this... Y-y-your question is just bad. Come on. Of course, I care about you. Of course, I'm gonna try to help you. You gotta stop doing what you're doing right here.

01:21:40,732 --> 01:21:40,872 [speaker_14]

You're ask-

01:21:40,912 --> 01:21:41,402 [speaker_0]

This is not good for you to watch.

01:21:41,402 --> 01:21:42,832 [speaker_14]

You're asking for an argument-

01:21:43,132 --> 01:21:43,142 [speaker_0]

If I were-

01:21:43,142 --> 01:21:45,712 [speaker_14]

... and then you're saying I'm embittering you for arguing with you.

01:21:45,832 --> 01:22:02,712 [speaker_0]

It's an ugly question. Let's try to reformulate. Eric, I have some concerns. I don't actually know all of your, uh, incentives. Uh, to those in our community who might worry that you're a self-promoting person, what would be the best way of answering them? That would be a good question. It's sort of pretty close to your question.

01:22:02,752 --> 01:22:03,512 [speaker_6]

Can you answer that?

01:22:03,732 --> 01:22:07,952 [speaker_14]

You're, you're imposing a power dynamic on our discussion. I thought we were coming into this equally.

01:22:09,972 --> 01:23:01,352 [speaker_0]

Do you wanna talk for five minutes? I'll listen. I'm doing a really difficult thing. So what I'm trying to say is, I wrote an article called-- on Russell conjugation for edge.org, and it said that there's a weird thing about the English language, that the emotive valence of a word and its content do not have to be the same thing. So whistleblower and fake are technically synonyms, but they are emotionally antonyms for the same meaning. The problem isn't in what you're asking. The problem is, is that you're not clocking a standard collegiality that-- You're, you're effectively insinuating that my motives aren't pure enough, that I don't wanna help you. And my feeling is: How do you wanna help me? Because this is teamwork. We're not gonna-

01:23:01,392 --> 01:23:02,612 [speaker_14]

So it does-- So it is transactional.

01:23:03,372 --> 01:23:35,552 [speaker_0]

No. No, no, no. Transactional-- What's the difference between dinner and a movie and street walking? It's layers of indirection. It's a question about is it a direct quid pro quo? Exchanges between people are not all transactional. But yes, I'm not offering to immolate myself and my credibility to have my colleagues who I hold up in highest esteem, uh, undercut me as soon as they get to Washington, DC. I played that game. I'm sixty years old. I'm not doing it again.

01:23:38,152 --> 01:24:02,362 [speaker_15]

Um, I'm curious about your choice of audience and choosing to direct the advice to scientists to advocate better for themselves rather than direct, you know, have your audience be DC and say, "There's a community that needs to have more attention and more funding and more respect." Like, what-- Is the onus on scientists to command that respect by-

01:24:02,392 --> 01:24:05,772 [speaker_0]

I told you I've addressed the National Academy of Sciences five times.

01:24:06,992 --> 01:24:08,892 [speaker_15]

Right. And what was the response?

01:24:09,712 --> 01:24:11,362 [speaker_3]

But it's all the scientists who are like us.

01:24:12,312 --> 01:24:23,182 [speaker_0]

No, no, no. Those people are functioning as policy administrators. Very often, they're hostile to scientists. Th-the-- You, you don't understand the Washington dynamic. I bring your case to the best of my ability. Every time-

01:24:23,192 --> 01:24:31,552 [speaker_15]

But it still, uh, I mean, it still seems like the suggestion to us is to play a dirty game of politics.

01:24:31,852 --> 01:24:33,072 [speaker_14]

No, just to be smart.

01:24:34,732 --> 01:24:37,562 [speaker_0]

Where did you get dirty game? You're funded-

01:24:37,562 --> 01:24:38,092 [speaker_15]

You said, you said that.

01:24:38,372 --> 01:25:41,580 [speaker_0]

Okay. The reason that you're funded by the federal government is the federal government, according to, to Weber in the theory of government, you may n-never heard this, the government is a monopoly on violence, okay? We got a technical legal. The reason we can collect taxes is that we can threaten to put people in jail, or we can put a lien on your property. So the monopoly on violence can extract money that nobody feels like paying at any given time to the, to the federal government, and then we give that money to you. That is a dirty game in a certain sense. We are not without sin. My claim is that we should be as ethical as possible in advocating for ourselves using truth as aggressively as possible. Truth does not have a particular narrative. It is up to us to advocate in an adversarial process for our slice of what we have provided to our country so that we can continue to do it or to it. I would love to help craft that narrative, but if you feel that it's-- you're sullied by it, I understand.

01:25:41,980 --> 01:25:42,360 [speaker_16]

Well, I don't-

01:25:42,440 --> 01:25:48,120 [speaker_0]

But then keep in mind that the way in which you're getting your money is by forcing people to pay taxes which no one wants.

01:25:50,480 --> 01:25:51,490 [speaker_17]

It was great. Thank you.

01:25:51,490 --> 01:25:53,540 [speaker_0]

You get to speak.

01:25:53,660 --> 01:27:42,500 [speaker_17]

Yep, yeah, yeah. Hi. Um, so I wanted, uh, to revisit, uh, like a point that that, um, lady, uh, said, like about, uh, the, the burden of proof. Um, so whenever a lab puts out a paper, usually the paper is highly criticized, like through an editorial process. And, uh, then like a, a discourse is had and, uh, they either like come to an agreement or, uh, there is some further work that like has to be done. And whenever a professor sh-sh-sh-sh-shocks or whoever has, uh, problems with your theory, it seems like either you don't want to answer the question because it opens up holes that you aren't, um, willing to rectify. Like yesterday or, um-

01:27:42,660 --> 01:27:42,970 [speaker_0]

Tuesday

01:27:42,970 --> 01:28:02,400 [speaker_17]

... on Tuesday, whenever like there was, uh, talk about the s-s-s-s-- the issue of s-s-s-s-s-scaling or of the graviton, graviton s-s-s-s-s-scattering-

01:28:02,660 --> 01:28:02,820 [speaker_0]

Yeah

01:28:02,900 --> 01:28:39,380 [speaker_17]

... it s-s-seems like you hid behind like technical debt and then moved on. And, and it isn't that like I don't have respect like for your work. It's just that like there seems to be issues, and you sh-sh-sh-should talk openly with like our community. And like I think that you-

01:28:39,460 --> 01:28:40,470 [speaker_0]

I think I get the gist of it.

01:28:40,500 --> 01:28:55,180 [speaker_17]

Right. But like I, I want to emphasize that it isn't like we hate you or it's like we don't want to talk.

01:28:55,240 --> 01:28:55,250 [speaker_0]

I do-

01:28:55,250 --> 01:28:59,500 [speaker_17]

It's that like you like say, "Technical."

01:29:01,300 --> 01:29:02,020 [speaker_0]

Let me be clear.

01:29:02,070 --> 01:29:06,960 [speaker_17]

"Debt," and then move on. But like why are we moving on? Let's, let's talk about it and figure it out.

01:29:07,100 --> 01:29:10,080 [speaker_0]

I told you I would love to come back for a week.

01:29:11,810 --> 01:29:11,810 [speaker_17]

Right.

01:29:11,860 --> 01:29:31,460 [speaker_0]

At my expense. The problem is why does it have a problem that a Harvard PhD in mathematics who's an MIT postdoc, why does he have the problem to get a visiting appointment on his own money in a department where he's clearly competent and great? Wait, just a second. I listened to you.

01:29:31,640 --> 01:29:32,150 [speaker_17]

Right, right.

01:29:32,400 --> 01:29:32,870 [speaker_0]

Hear, hear me out.

01:29:32,880 --> 01:29:33,280 [speaker_17]

Okay.

01:29:34,780 --> 01:30:12,880 [speaker_0]

There's a lot you don't see. So nobody's recording this, let me just say some stuff. I don't like being doxxed and having my personal address broadcast on the internet. There's a Discord group filled with PhDs who are technically critic, critics of mine. And you see that, "Why won't Eric answer a question about chiral anomaly?" And below the surface, you don't see, "Should we help a mentally unbalanced person find Eric's home address? Why should we, uh, why do we care about any Eric for any reason other than entertainment?" Or, "Didn't I hear that Eric selectively likes to molest little girls?"

01:30:14,110 --> 01:30:14,110 [speaker_17]

What?

01:30:14,140 --> 01:35:04,392 [speaker_0]

Yeah, I'm telling you. I'm telling you. It's Scott Aaronson who brought this up. There are psychopaths who are podcast guests who write to my wife's advisor, who try to dig up, turn my friends against me. You're, you're talking about troglodytes and celery farmers that you have chosen to normalize, that I choose not to talk about because I go to Gavin de Becker, and he says, "You've got a real stalking problem." Right? So my point is I'm dealing with problems you can't even imagine. I, I announced the time and the place. I don't know whether I've got a mentally unbalanced person like the person in Colorado who killed five people in, in tattoo parlors who was trying to get my attention. The FBI calls me. I'm in a position, I'm in positions you don't even know about. Okay? I only announced this a couple days ahead of time because I don't want to give everybody in the country an opportunity to buy a ticket and get to Austin. I don't know what I'm doing. You have no clue what I'm up against or what I'm doing. I've given you the answer, which is this is what comes out and emerges from large things, from large periods of time. One of the things that is e- very bittering is, is that you don't know all the dirty games. Like, there was a question about the purity of the field. I didn't know there were secret seminars where we decide who's in and who's out, who gets the information, and who doesn't. I didn't know that there are, like, campaigns to rename something, uh, based on somebody does

the work and somebody else needs it named after. Like, there's an entire subterranean world of academics that I'm trying to navigate outside defenseless as a single individual. So I'm gonna take some countermeasures. But if what you wanna do is to talk science, and you-you've got a, an open heart and an open mind, I'm very interested in those questions. But if you ask somebody-- Like, I know your community well enough that if you don't wanna listen to somebody, you do the following thing. "Well, your theory is very interesting, but, uh, I'm only really gonna take it seriously if you can show to me the ratio of the neutron to proton in the mass of the first proton." Were you? Where did you come up with that? Did you, did you ask that to Andy Strominger? I mean, you guys don't realize how transparent some of the weird games that take place inside the community, that do the community no service, are to people who are informed about what's going on. There are all sorts of questions that we ask to get some crackpot to leave. Come back when you can ask. And we don't ask that to a colleague. So for example, when Ed Witten presented at MIT the theory that would generate what is now called the Seiberg-Witten equations, which I came up with in nineteen eighty-seven, he didn't actually write down the equations for the talk. A guy named Alan Knutson at the end, in the question and answer, said, "You say that they're simplified equations that generate all of Donaldson theory, but you didn't write them down." He writes them down. And I said, "Oh my gosh, he-- I think he used phi where I use psi 'cause he was s- s- spinning one-half through." I said, "My, my God, that-- those are my equations." Cliff Taubes is gonna tell me, um, "Well, those are insufficiently nonlinear, and spinors have nothing to do with Donaldson theory." There's no paper. There's no peer review. The field was revolutionized in a few weeks because of a talk that Ed Witten gave. Your selective application of your standards is a glaring lever that you use to get rid of your crackpots and crazies, right? The idea is you can always apply them when somebody feels like, "Okay, well, I, I guess I got my technical detail." But you supp- apply them selectively. And what's bizarre is it's a little bit like when you have a conversation around me in Turkish. I actually speak a little Turkish, you know? And so in a weird sense, you're not expecting that I might catch these. You know, you call me strange. You know? If you're in Hindi, you might say pagol. Oh, I'm pagol. You're like, "Oh, the white guy knows what pagol means?" You don't realize that you're leaking. And we've had several leaky questions here. So what I'm gonna say back to you is very fair point. But if we hang out together, you won't have the same attitude. You'll have me saying, "Actually, I don't know that. I didn't take a course in quantum field theory. I, I went one semester of classical mechanics. That's it." So, you know, very often, um, I'm an eager learner. So thanks.

01:35:05,132 --> 01:35:07,652 [speaker_3]

Sounds like you have a follow-up right next to you.

01:35:07,772 --> 01:35:33,611 [speaker_18]

Uh, I was just gonna ask, like, you had said, you know, that you were gonna give yourself a commitment to the scientific community and, um, but in the sense of, like, offering, uh, advice for, like, how scientists should be operating in sort of like, um, economic fields that we're in now, uh, you had mentioned that, you know, taxes from the people are bad. Is this correct?

01:35:33,812 --> 01:35:34,852 [speaker_0]

No, I didn't say they're bad.

01:35:35,272 --> 01:35:36,642 [speaker_18]

Or what you said is, uh-

01:35:36,671 --> 01:35:37,702 [speaker_0]

They're extracted through violence.

01:35:37,702 --> 01:35:40,482 [speaker_18]

Yeah, through violence. Um, and-

01:35:40,632 --> 01:35:41,332 [speaker_0]

Which I'm for.

01:35:42,732 --> 01:35:43,802 [speaker_18]

No, I'm not kidding. I-

01:35:44,032 --> 01:35:45,752 [speaker_0]

We haven't figured out a better system.

01:35:45,782 --> 01:36:01,092 [speaker_18]

That's true, yeah. But, um, most scientists, um, like especially in academia, are getting their funding through, you know, federal avenues. And, uh, would you recommend, I guess the, the point of this all is would you recommend, you know, moving to the private sector-

01:36:01,172 --> 01:36:01,392 [speaker_0]

No

01:36:01,452 --> 01:36:02,512 [speaker_18]

... to fund your facility?

01:36:02,552 --> 01:36:20,032 [speaker_0]

No, we gotta get either federal money again. The federal government is the right payer. And don't get moved on GoFundMe, where you take an actual claim to the billions you're owed and exchange it for seventeen thousand dollars from some friends.

01:36:22,052 --> 01:36:38,332 [speaker_18]

But if, uh, so if the goal is still to look for funding through federal avenues, then are, like, with the dependence on federal money on the administration or the current political climate, then what is the best, like, uh-

01:36:38,612 --> 01:36:39,852 [speaker_0]

How do you appeal to the Trumpies?

01:36:40,732 --> 01:36:42,122 [speaker_18]

Or maybe not even just them, but-

01:36:42,152 --> 01:36:43,412 [speaker_0]

You're asking me how to be a parent

01:36:43,532 --> 01:36:49,802 [speaker_18]

... whoever comes next or the administration after that. Sure, they're Trumpies now, but you never know where the winds will blow us.

01:36:50,172 --> 01:39:22,836 [speaker_0]

Okay. I highly recommend reading the nineteen ninety-three hearings. As many of you know, um, the claim that correlation does not imply causation. But when Jacque pissed and moved to Texas, the, uh, Superconducting Super Collider was canceled within a week. [audience laughing] And, um, so, so great was the fear of the Canadian genius that, uh, we d- we, we dare not put it close to him. The hearings in which That was canceled should be read by all of you. That was a watershed moment, um, for the community. It was sort of the death of federal commitment. You guys used to have carte blanche because you were the bomb guys. And what

happened was that there were some, like, there were some art purchases for the SSC, um, and people started, you know, acting like BSDs, uh, with this federal funding to put this collider into the ground, and there was a lot of anger. So there, uh, Representative Walberg, I believe, was on a hearing. There were these opening statements where they said, "You know, we're really disappointed in the science community because we have trusted you for ages." And the two things that broke trust were how the SSC was handled and how the science shortage took over Washington. The Immigration Act was passed, and then there wasn't a science shortage. So you, you suffered a catastrophic, catastrophic loss of the carte blanche and bipartisan support around that time, and it's worth reading the opening statements of the representatives who were most livid and angry. And when you piss off somebody in Congress, a single person can do fantastic damage. You're supposed to have representatives who are smoothing those feathers because, like, Texas was gonna get this benefit of the SSC. So you had, well, what does New York get? And you can say, "I don't wanna play the dirty games," and people just left. Okay, do you want federal funding at all? My recommendation is do what the people in Hollywood do. If you're gonna live with somebody on set in trailers, you don't want them doing the negotiation. You have an agent who's a son of a bitch. You need your own sons of bitches. They don't have to be you, but they're supposed to be smart. They're supposed to have your support, and you have to recognize that you need a layer of indirection between the beauty of the science community and that which makes the sausage.

01:39:23,596 --> 01:39:25,976 [speaker_3]

Let's quickly get John and then Besson.

01:39:26,036 --> 01:39:30,096 [speaker_11]

Okay. So, uh, just a, a, a couple of things to clarify. Um, uh, uh-

01:39:30,136 --> 01:39:30,456 [speaker_0]

Sorry, but-

01:39:30,556 --> 01:39:33,485 [speaker_11]

I arrived here after the SSC was canceled.

01:39:33,825 --> 01:39:33,825 [speaker_0]

Okay.

01:39:33,836 --> 01:39:36,276 [speaker_11]

But I did have the surreal experience-

01:39:36,316 --> 01:39:36,666 [speaker_0]

Representative Walberg

01:39:36,696 --> 01:41:33,476 [speaker_11]

... of, of turning on C-SPAN late at night and watching my future boss testify before Congress. Uh, but it was not just him, and, and this I think is actually an important lesson for, for all of the people in this room. Uh, the other person I watched testify before Congress, uh, was the guy who at the time had the office next door to mine at Princeton, namely Phil Anderson. And Phil's, uh, uh, uh, position, uh, was that the condensed matter community could do so much more with the nine billion dollars or whatever it was that the SSC would cost, um, and that, um, therefore, the money should be spent not on the SSC, but elsewhere. And, and what he failed to understand is that this is not a zero-sum game, that the money that didn't get spent on the SSC was not gonna get spent on condensed matter physics. It just disappeared back into the federal budget. Uh, and, and, and that's a really important lesson for, for all of us that it's not a zero-sum game. And, and, and this, um, to my, uh, to my great dismay was, was Phil

Anderson's perhaps greatest mistake. Um, the thing I wanted to ask about, uh, um, because it seems to me in terms of the sort of both the power dynamics and, uh, what's changed part is the slide you had about, uh, uh, uh, your cohort in, in Silicon Valley and how their view of science seems to have changed rather radically. And given the influence that they have in Washington and will continue to have, um, what can we do-- Well, what is it that changed their mind, and what can we do to, uh, hopefully change it back? Because that I think is-- we're not gonna get very far unless we do that.

01:41:34,536 --> 01:41:50,296 [speaker_0]

Great question. [clears throat] First of all, the most honest thing to say in a science-- to a science audience is I don't know. I got ideas. Roughly speaking, I go to seminars. I go to seminars at Caltech and UCLA more than I go anywhere else.

01:41:50,636 --> 01:41:50,776 [speaker_11]

Mm-hmm.

01:41:51,066 --> 01:45:16,568 [speaker_0]

And I give seminars. And in general, the people who are saying universities are over and that science is over, thank you, um, don't go to seminars. Nobody is coming to our seminars. They can't see when somebody's full of shit. They can't see when somebody is killing it. They just don't know. And what I did by bringing two badly behaved comedians and, and a third podcaster is to try to open the lines of communication because you have something that is nothing if not entertaining. So I'm just gonna make a point that's adjacent to this. Nobody says, "What is the Hubble telescope good for?" They just say, "Whoa. That telescope's unbelievable." [laughs] Right? Because it's immediate, right? You don't have to sell sugar to children. Eat your sugar. No, no, no, you have to, you have to eat your Brussels sprouts. So my claim is what do you have that makes them feel good, feel smart, understand what you're doing? Like, I know a lot of physics. I do not grasp what is going on in dimensions we don't live in, in toy models that aren't our world forty-two years later. I just don't get it. I've sat there for the whole thing. It, it doesn't make sense. So you can try to say, "How would I explain that to Marc Andreessen?" Who's now become one of your biggest problems. But my claim is if you can explain it to me, I can try to explain it to Mark Andreessen, and I don't get it. So partially, here's the thing. We have to cut some of the stupid shit we do loose. We can't keep it all. Some of it is just dumb and past its sell-by date. Not all members of the community need to be in science. There are some people who need to be doing other things with their lives. That said, there is a giant core of real stuff that we do that is essential, that is done nowhere else, that cannot be supported by the private sector. So it's a multi-pronged thing. One, we have to say, you're right, we should not have had seventy-seven Nobel Prize winners writing in support of the EcoHealth Alliance and its restoration of gain of function, whatever. Where, where is that? We lost control of an area. I'm gonna say something really tough to hear. The so-called IC, or intelligence community, is in opposition to you guys at an incredible level now. Why? Because you can tell what's true and what's false. I right now am listening to stuff on Fox News about four species of aliens piloting craft that are invading our airspace, and my point is, give me one square millimeter of tissue, and it is gonna be over in forty-five minutes. Are you gonna tell me it's eukaryotic? Are you gonna tell me it's carbon-based? Are you gonna-- Nobody can fake cytology and histology, okay? So if-- imagine that that's a bullshit campaign by the IC. We are their worst nightmare. What happened in Los Alamos? How many-- Who knows what the Reference Committee is? What's the Reference Committee?

01:45:16,628 --> 01:45:18,568 [speaker_1]

Oh, uh, what happened in Los Alamos?

01:45:19,308 --> 01:45:20,388 [speaker_0]

What is the Reference Committee?

01:45:21,348 --> 01:45:28,628 [speaker_1]

Um, I don't know about the Reference Committee, but I, uh, thought you were referring to something else that happened in Los Alamos a few years back that I'm familiar with.

01:45:29,268 --> 01:49:30,092 [speaker_0]

The Reference Committee, I believe, was the thing that was set up inside the National Research Council, which is the working part of the National Academy of Science complex, that said if you submitted a paper on chain reactions, it didn't just go to the journal, it went off to the Reference Committee, which I think was chaired by Bright, uh, Bright Wigner. And it was-- it would be sent back for revisions because they wanted to slow, slow acceptance so that you get-- got so frustrated, you changed your, your focus and your research. So we got chain reaction physics to die outside of a boys' school in New Mexico. The IC functions in a way in relation to you that you need to be cognizant of. You need to know that Elaine Maxwell's father destroyed scientific publishing with Pergamon Press, and he did it by ballooning the number of journals because every university needed to have a complete collection of journals, so as many journals as he could produce, everybody had to buy all of them. Genius idea. That's why Elsevier is so profitable. It's also how we control scientific information. What we do is dangerous. Teller-Ulam design is classified. John Aristotle Phillips submitted a junior thesis at Princeton to-- with Freeman Dyson as his advisor, which has been taken out of the library, and page twenty is not available because it's a working model of a fission device. He was the mascot for Princeton, and he said, "I'm a bad physics guy. Let's see if we can use my bad-- being bad at physics to show that there is no secret to the atomic bomb." Freeman Dyson said, "Let's do it. I will give you no information." What we do is dangerous. What we do has to do with the Department of Energy. Department of Energy is really the Department of Physics, right? We had the AEC, Atomic Energy Commission, which is really-- Atomic energy is a euphemism for atomic weapons, which is a euphemism for physics. The Department of Energy has counterintelligence units. It's part of the IC. What we do is dangerous. The IC is worried in part because you guys can tell truth from fiction. You have instruments, you have brains, and the government lies at a very low level all the time. So in part, you need liaisons into the CIA Science and Technology group, into the Department of Energy's counterintelligence wing. You need to be part of the security conversation of the United States of America, and you're not. In fact, you treat it as if, well, this is the province of Alex Jones. I don't know what to tell you. Alex Jones isn't wrong about everything just because he's Alex Jones. This is really dangerous stuff. And by the way, for those of you who know my Jeffrey Epstein story, Jeffrey Epstein was very focused on gravity. He was very focused on buying Zorro Ranch because it was near Sandia and Los Alamos. It was a listening post. We do not know what Jeffrey Epstein was up to. But my claim is, is, is that any rich weirdo who buys a ranch to be close to these things should be under direct inspection of the Department of Energy, which is really the Department of Physics. So what I'm trying to say to you is, you need a more sophisticated picture of-- Registered conspiracies are called special access programs. Unacknowledged ones are USEPs. Waved and digged is, is WABUSEP. These things are real. They involve physics. You need to be part of your own conversation. None of you are probably part of the JASON. You can't tell me if you are. But you don't realize how important these advisory services are and these linkages, and some of you need to be going to Mar-a-Lago no matter how much you hate Donald Trump. Last question.

01:49:30,852 --> 01:50:48,992 [speaker_1]

Right. So I kind of feel like throughout this discussion, we've kind of been dancing around the, like, the real base level issue here, which is why we've seen such catastrophic cuts, uh, to funding, including, you know, early, uh, an illegal impoundment early in the administration. Um, and this has largely been as a result of the, uh, perception among, uh, among conservatives, among people on the right side of the political aisle, that universities are supposedly dens of,

like, woke transgender Marxists. They bring up things like DEI, which-- And I will, I will gladly express, like, some criticisms of diversity, equity, and inclusion initiatives, but they, they characterize this as including basically any effort to, you know, make the university a, a hospitable space for women, for transgender people, for, uh, the disabled. I, I especially talk about, like, transgender people because this seems to be a big obsession, um-

01:50:49,252 --> 01:50:49,632 [speaker_0]

Huge.

01:50:49,852 --> 01:52:25,572 [speaker_1]

Yes, a huge obsession by the political right about a group of people that is largely pretty small and contains a lot of our elite human capital. Now, why would, why would anyone who presumes themselves, you know, a supporter of, you know, the sciences-- I'm not saying this of you, I'm saying this of policymakers generally. Why, why would they, you know, why would they pursue-- Why would they actively pursue policies that are meant to drum, uh, our practically elite human capital out of the university system simply because they view them as, like, you know, they see the, they see the fact that they're transgender. They see the fact that they have, you know, le-left-wing social views. They see the fact that they, uh, that, that, that they disagree with, uh, you know, American, um, foreign policy of, of, uh, of arming Israel unconditionally, and they want to, like, you know, enact revenge against them. Like, this is fundamentally the base level concern of why, you know, this, like, it is especially true as of late that the right has gone on this all-out attack against all of academia. And I feel like this hasn't been mentioned more or less at all, you know, by anyone. Not-- It's not just you, it's any-- almost anyone else explicitly in this two-hour talk. And I, I want to ask, like...

01:52:27,592 --> 01:52:28,152 [speaker_0]

Great question.

01:52:28,232 --> 01:52:28,512 [speaker_1]

Yeah.

01:52:29,152 --> 01:52:29,552 [speaker_0]

You want me to answer it?

01:52:30,292 --> 01:52:30,572 [speaker_1]

Yes.

01:52:32,492 --> 01:54:57,412 [speaker_0]

Okay. So just a little bit of personal history. Not that I want to share it with you, but it will inform the answer. Long before trans was a word in all of our vocabularies, I had an issue called intersex, which is that, as I said, I'm a little Turkish. In Turkish, the pronoun third person singular is o. It is not inflected for gender. And just as marital status for men was mister and for women it conveyed more information, I thought that there was a one-time change that one could make to create a single pronoun that hid this in the English language. I've talked to linguists about all these things. So trans wasn't my issue, but intersex was very much my issue. And it-- most of the work for trans can be done by intersex. They're not angry about trans. They start off being angry about not being able to say simple things. In other words, and this is very important, there is gender and there is sex, and they, they were actually pretty much the same thing until the fifties, and then we decided that we wanted to use one word for one thing, one word for something else. Normal expectations of reproductive gender and sex were largely things about bundles back in the fifties. A bundle of characteristics. In India, I wear a lungi. In the US, I would be a dude wearing a skirt. I wear a lungi. It's okay. The issue about how we dress, how we behave, how we talk, all these things, is not really the issue. The issue is when

you can't say things, when you tell somebody, "You can't say X." Right? The N-word is particularly potent among ethnic slurs. With a hard R, we have endless discussions. What they are fed up with is they start off being fed up with being told that they are bad people for trying to say something. And we-- they want to say something about a swimmer based on human sexual dimorphism between sexes. Wait, let me get to the end.

01:54:57,472 --> 01:54:57,732 [speaker_1]

Oh.

01:55:00,672 --> 01:56:42,208 [speaker_0]

These things are so divisive when we start talking about them, and we don't have the discipline and the decency. But if you look at, for example, the distinguished economist, Dieter Dreher-McCluskey, who used to be Donald McCluskey, there is not a human cry in the economics profession about Dieter. Just isn't. And I talked to her, and what she said to me was, "I didn't want to be sexy or hot. I just wanted to die as a grandma." And I don't think that most people in the administration have any problem with different philosophy. Trying to figure out whether they're threatened men who are having a masculinity crisis, or whether they're actually reasonable people who started out saying, "I-- why can't I say something that is true about allowing men and women to compete in MMA based on sexual dimorphism?" A lot of this comes about by saying, "Don't say X. Don't say that vaccines have any problems. Don't say that this might have come out of a lab. Don't say that there are sexual dimorphism issues between males and females. Don't say that there may be gender dysphoria that has to do with mental instability." Okay, so now you're, you're trying to pre-game all sorts of things that normally would have been a part of science because of the so you have a look at function. This is going to end badly. So what I would say is, yeah, a lot of them are just being dicks, to be honest. But a lot of it came from the universities being scolds. Quite honestly, it's a messy, bad conversation between two groups that should be having a better conversation.

01:56:42,288 --> 01:56:43,208 [speaker_1]

Okay. I-

01:56:44,328 --> 01:56:44,508 [speaker_0]

Yeah.

01:56:44,978 --> 01:57:12,768 [speaker_1]

Yeah. So I have a few things to say about this because I kind of feel this is a very, very common sort of folk history of events that, uh, honestly is not really accurate. First of all, um, you know, you started off mentioning that our conception of, like, sex and gender was basically the same thing until the fifties. This is a common myth. The term gender actually was, uh, first used to-

01:57:12,808 --> 01:57:12,958 [speaker_0]

Grammatical

01:57:12,958 --> 01:57:17,188 [speaker_1]

...refer to, yes, grammatical gender. Um, like this, this-

01:57:17,298 --> 01:57:21,288 [speaker_0]

I'm being-- this is... Are we sure that we wanna turn to general talk? You wanted to-

01:57:21,458 --> 01:58:07,698 [speaker_1]

This, this is, this is a small point, but I, I, I, I, I, uh, I did not want to, like, separate on it. What I wanted, what I wanted to focus on, because I, I think I found this to be the particular, like, point of, uh, you know, the, the real point of contention here. It's, it's this idea that, like, the culture

war, if you will, was ignited by a refusal by, you know, one side, uh, denoted as being like the universities, you know, advocates for the rights of transgender people, et cetera, et cetera, et cetera, to allow, you know, uh, dissenting opinions. This is, uh, this is quite sincerely balderdash. If you look at the discussions around-

01:58:07,718 --> 01:58:08,408 [speaker_0]

Can I say something?

01:58:08,698 --> 01:58:49,088 [speaker_1]

If, if you look at the dis... I, I, I, I, I, uh... I, I, I waited, and I would like to, I'd like to, you know, I'd like to be able to make my, my, my point and reply. Um, the, uh, the discussions around, you know, transgender people around ten, twelve years ago, um, particularly from, like, the cultural right, they were, um, full of, uh, you know, d-derision and invective and, you know, like, s-scare tactics about, "Oh, they're coming for your children. They're going to go into, into, you know, women's bathrooms and-" This was a conversation about physics. Yes. Yes. So can we get to the physics?

01:58:49,148 --> 01:58:50,708 [speaker_0]

I feel we're getting off track.

01:58:50,737 --> 01:59:09,408 [speaker_1]

Yes. Yes. I, I, I, I, I, I do want to say, like, this, uh, this idea that, like, everything was okay until one side, you know, refused to allow the other side to talk is simply, like, not an accurate telling of the, of events.

01:59:09,438 --> 01:59:09,438 [speaker_0]

I have two-

01:59:09,448 --> 01:59:12,928 [speaker_1]

And I think, and, and I think that this is relevant to the conversation-

01:59:13,068 --> 01:59:20,517 [speaker_0]

Part of what I'm gonna do is I'm gonna scope this thing. No offense, but what I'm gonna say is I didn't give a land acknowledgment. I'm sure that this land was held by other people. There are a bunch of other issues here.

01:59:20,528 --> 01:59:20,648 [speaker_1]

Yes.

01:59:20,818 --> 01:59:21,528 [speaker_0]

It's a real issue.

01:59:21,548 --> 01:59:22,138 [speaker_1]

Somehow, uh, whatever.

01:59:22,138 --> 02:00:30,728 [speaker_0]

It's a real issue. And then, you know, there are real people here who are in danger of being deported under the new zealously of the administration who deserve to be here. I can't tell you how many issues I see and feel personally. Part of what we do here is that we scope things. So I'm going to say if the transgender community of the University of Te-Texas at Austin wishes to have me back for a discussion about the human capital lost to bigotry from cultural conservatives and a zealously bordering on mania, I am interested in having that conversation, and I will not allow a discussion of indigenous land rights to invade that

conversation. If the First Peoples wish to have me talk about whether or not they're being properly included or the entire waste of Black capital in the United States, so which is not entering a correct pro rata share of the science, uh, of the STEM enterprise, I'm interested in all those conversations. I gladly will return.

02:00:30,928 --> 02:00:32,088 [speaker_1]

I, I, I have a point about-

02:00:32,348 --> 02:00:32,358 [speaker_0]

That's it.

02:00:32,388 --> 02:00:35,428 [speaker_1]

I have a point about the, the main subject of physics that I'd like to bring up.

02:00:37,408 --> 02:00:39,808 [speaker_0]

That's it. That's it.

02:00:39,888 --> 02:00:40,308 [speaker_1]

My, my point-

02:00:40,388 --> 02:00:40,418 [speaker_0]

That's it.

02:00:40,418 --> 02:00:40,418 [speaker_1]

My point-

02:00:40,418 --> 02:00:40,588 [speaker_0]

That's-

02:00:40,678 --> 02:00:41,377 [speaker_1]

...was that with your-

02:00:41,377 --> 02:00:41,768 [speaker_0]

That is it

02:00:41,778 --> 02:00:52,377 [speaker_1]

...GU theory, you talked about how you were being silenced by, you know, like physics departments and Sean Carroll and physics influencers and such. But really what was happening-

02:00:52,448 --> 02:00:52,658 [speaker_0]

That is it

02:00:52,658 --> 02:00:54,628 [speaker_1]

...was you were experiencing pushback-

02:00:54,828 --> 02:00:55,568 [speaker_0]

That is it

02:00:55,608 --> 02:01:02,028 [speaker_1]

...for, you know, like your, your ideas were being challenged and criticized. This is fundamentally the same thing.

02:01:02,048 --> 02:01:09,128 [speaker_0]

The failure to read what is happening in this conversation has nothing to do with transgender people.

02:01:09,188 --> 02:01:10,168 [speaker_3]

Can we move to the next?

02:01:11,608 --> 02:01:54,908 [speaker_19]

Yeah. Hi. Um, I am not in academia anymore, thank God. Um, but I do a lot of independent research. The first thing when it comes to going farther in things that I don't quite understand is I reach out to people. I have no shame in doing that. Um, but I do have a major fear of having my business and my research stolen from me. For example, I researched a certain professor at a certain university recently, and eight days later, there was a paper out. So that was a lot of fun. Um, right on to our next. So how do, how can-- and I also hire people from the faculty. So like how can I self-engage like in-

02:01:54,998 --> 02:01:56,458 [speaker_0]

It's an unsolved problem.

02:01:56,738 --> 02:01:56,748 [speaker_19]

In-

02:01:56,758 --> 02:02:12,358 [speaker_0]

And I wanna talk about why it even exists. In general, when two academics greet each other, it's a lot like two dogs sniffing each other. The immediate question is, what field are you in? What subfield are you in? What is your title? What is your rank? What is your position? Right?

02:02:12,368 --> 02:02:12,978 [speaker_19]

That's fine. I'll probably do that-

02:02:13,038 --> 02:02:16,038 [speaker_0]

Why do we do-- Why, but why do we do that? No, no, to your question.

02:02:16,398 --> 02:02:16,678 [speaker_19]

Yeah.

02:02:16,698 --> 02:02:33,538 [speaker_0]

We do that because we're trying to figure out-- not that we know that we're doing this. We're trying to figure out if we're in repeat or single games. When you come in from outside, you effectively are like a hot chick getting off the bus at Hollywood & Vine-

02:02:34,098 --> 02:02:34,108 [speaker_20]

What?

02:02:34,108 --> 02:02:36,298 [speaker_0]

-hoping to make it in Hollywood.

02:02:36,438 --> 02:02:36,847 [speaker_20]

What? What?

02:02:36,847 --> 02:02:43,898 [speaker_0]

What I'm trying to say to you-- I have this problem too, by the way, as you can see. Um, once it's a certain-

02:02:44,538 --> 02:02:45,108 [speaker_20]

I know this is not supposed to be funny

02:02:45,108 --> 02:03:00,128 [speaker_0]

... it's, it's kind of an independent researcher. Like, my esteem for you just went down for reasons that I can't quite explain. I'm being, being honest with you, which is not something we typically do. You're more likely to be a crank. Again, it's a probability, just I'm not saying you are a crank. But my point is-

02:03:00,158 --> 02:03:00,648 [speaker_19]

Well, that's-

02:03:00,718 --> 02:03:00,728 [speaker_20]

What?

02:03:00,728 --> 02:03:18,578 [speaker_0]

-is that when you say, "I'm an independent researcher, and I have concerns that people can steal my work," it's like, "Oh, here's the bell. Here's the bell-shaped distribution." I have an academic professor in a sandbox in my mind, and I run an emulation. So you just lowered your status, and you told me your vulnerability.

02:03:19,958 --> 02:03:27,148 [speaker_19]

Well, that's my question is how can the outsiders approach individuals and with-

02:03:28,078 --> 02:03:28,718 [speaker_0]

Do you see that-

02:03:29,128 --> 02:03:29,548 [speaker_19]

-without this fear game?

02:03:29,558 --> 02:03:45,158 [speaker_0]

I would go to the top of my, go to the top of my feed. That tweet tweeted out by Joe Rogan and Sean Carroll is what would be called technically the return of costs. You have to return costs as a lone wolf.

02:03:47,278 --> 02:03:48,938 [speaker_19]

What would academics find valuable?

02:03:49,718 --> 02:04:57,238 [speaker_0]

My point being that once it is known that somebody doesn't have, they suffer in the community. So that we can talk about all sorts of techniques. I can say to you that there is no solution to this problem. But I really do wish to tell you something that no one will say to you. The odds of you doing real things as an individual researcher plummet because of the cost of isolation. Mo-- If you look at me, look at my condition. Sixty years old, haven't been an academic for a long time, many haters. That should shift your probability estimation if you're honest. What's more, the odds of screwing me over and getting away with it are quite high. So in part, what you have to do is you have to take a dispassionate look as to why your colleagues are so terribly behaved. And in part, the reason that they're really terribly behaved is is that they're desperate. So you have to have compassion for the abusive people in academics who savage you because they do not have better options as they speak.

02:04:57,778 --> 02:05:01,087 [speaker_19]

For me, I just wanted dialogue, not just, you know, a post.

02:05:01,348 --> 02:05:03,298 [speaker_0]

I'm just giving you a very honest answer.

02:05:03,398 --> 02:05:03,428 [speaker_19]

Yeah, yeah, of course.

02:05:03,428 --> 02:05:05,098 [speaker_0]

You have an unsolved problem.

02:05:05,138 --> 02:05:05,518 [speaker_19]

Yeah.

02:05:06,858 --> 02:05:15,318 [speaker_20]

Yeah. Yes, obviously. Do you have a microphone? Yeah. I, I wanted to follow up. You were responding to the gentleman in front regarding AI.

02:05:15,698 --> 02:05:15,908 [speaker_0]

Mm-hmm.

02:05:15,908 --> 02:05:28,278 [speaker_20]

I thought this is sort of an interesting example space because we are all being taken from in the AI sense, and you said pose is a good way of how we should try to conduct to get our-

02:05:28,458 --> 02:05:28,838 [speaker_0]

Best start.

02:05:29,498 --> 02:05:46,158 [speaker_20]

The best one you've found to, to get the, to get the proper value for what's being sort of taken from us at scale. You have an interesting case where you can isolate what's showing up in the LLMs to be basically just you. So you have-

02:05:46,178 --> 02:05:49,458 [speaker_0]

All my haters. Like, I can tell which hater voices these are.

02:05:50,418 --> 02:06:03,478 [speaker_20]

So you, you have this interesting case where you can isolate it down. Have you tried to go to the, the Sam Altmans and the Elon Musks to try and get compensation-

02:06:03,998 --> 02:06:04,098 [speaker_0]

No

02:06:04,138 --> 02:06:07,098 [speaker_20]

... formal ones for that at all? Um, you haven't-

02:06:07,158 --> 02:06:07,598 [speaker_0]

I could.

02:06:08,678 --> 02:06:10,258 [speaker_20]

You, you think it would be a very good situation?

02:06:10,268 --> 02:09:39,554 [speaker_0]

I could get paid. I think I could get paid money. I am more interested in what we do. Look, how many of you have flown private? No, no, I'm not rich guys. Um, nothing special happened to fly

private. How many of you have been to, you know, top Michelin restaurants? Same. How many of you had an amazing bottle of Sutterra? It's not that great. [laughing] What we do is that great. So my claim is, yeah, I can get paid. I'm okay enough on my... It's not a great situation. It's not terrible. What is-- What matters to me is the health of our field and ultimately the question about whether s- whether it's possible to jailbreak general relativity and standard models. Because with the limitations that we know, we are on a very crowded planet that we have suffused with weapons and toys of incredible power. And given, given Xi, Modi, Putin, Trump, Netanyahu, and others, I don't see this ending well. We've had a remarkable period of eighty years of a very cryptic system holding the world together, including the petrodollar. Whooped that in nineteen seventy-four. It's unraveled. And the Trumpies are just so angry that they're blowing up all sorts of stuff that they don't understand. And the counter Trumpies were not maintaining it correctly either, so they emitted a lot of people. So you have the plague on both your houses. I don't care. I care that we, you know, everybody should read the letter between Ziller and Teller, where Teller tells Ziller why he's not going to collaborate with him on the ability to contain nuclear weapons. It's one of the most beautiful, pressing human things I've ever read. Like, we don't think of Teller as having a heart, but he's probably the only guy who really got it and said, "The only hope that we have is to make the war un-winnable." And we, the community that brought nuclear fire to the American political process and handed Truman these abilities, are complicit. We have blood on our hands, and we have to figure out whether interstellar travel, which is the only way to accumulate more than two spheres, the moon and Mars, which is what Elon... By the way, Elon was invited to come. He said, "I've got a lot of stuff on my plate today." We have to figure out whether we can make the science fiction needed for res- the restoration of an indefinite human future possible. That is our charge, so that it's not a punchline or a snicker. But the question is, do we really have a speed of light problem? Is there really no way that we can get anywhere quickly? Is there no way of jailbreaking space-time? Do we really live in space-time at all? My, my hope was that we would come together as a community and try to solve humanity's problems that we've learned to have created. Um, yeah, I could get my through this. I, I dictate a lot and shut up. But I'll do that if you guys are, like, representative of the disciplines in this program. I'd really rather save my, my clip.

02:09:40,654 --> 02:09:45,054 [speaker_21]

I was more asking the question from the standpoint of is there any advice-

02:09:45,094 --> 02:09:47,634 [speaker_0]

Individually? No. You should function collectively.

02:09:47,894 --> 02:09:48,894 [speaker_21]

Right.

02:09:49,674 --> 02:09:50,434 [speaker_0]

That's right.

02:09:52,474 --> 02:10:00,454 [speaker_21]

So when it comes to these technical things like those, like replacing the index numbers that we currently have with higher resolution to-

02:10:00,514 --> 02:10:01,754 [speaker_0]

In the economic sense?

02:10:01,854 --> 02:10:04,534 [speaker_21]

Yes. But actually, all these things we discussed.

02:10:04,743 --> 02:10:04,894 [speaker_0]

Okay.

02:10:04,934 --> 02:10:35,754 [speaker_21]

Um, it seems like we have a group of people that is very small, that are capable of understanding these ideas, but are totally incapable of working towards their implementation. You have a set of people that are also very small, that are capable of working towards their implementation, but are largely incapable or unwilling to understand them. And then you have an enormous swath of humanity that really impedes both of those groups, that are capable of neither. And so I'm a fan of these ideas. I would sincerely like to know how I can help, but I frankly feel a bit helpless.

02:10:36,294 --> 02:14:27,086 [speaker_0]

It's a good question. Dick Feynman on the Challenger com- on the, the Rogers Commission was a marvelous example of the danger of letting a scientist near this process. We don't control them. Once you have scientists asking questions and told, "You have a free mandate to sniff around and say what you find," um, the general, Kutina, I believe, got him the information he needed about the O-ring resiliency, and Dick Feynman dutifully played the genius that the general was, like, feeding him his lines. Um, and I believe that Neil Armstrong and, and, and Rogers met in the men's room and were overheard saying that, "That Feynman's a real pain in the ass." Yeah, that's what we are. We're real Fyn- pains in the asses. You can't control us. You can't tell us what to do exactly. We are... If there's Hitler to be defeated, maybe you can get a bunch of them to keep their mouth shut for a pretty great period of time. Part of what we do is we speak the truth when nobody wants to hear it, and we're very, very good at it. And what we-- what's happened is, is that everybody can see that... By the way, I don't know if you guys know what he's referring to with index numbers. The Boskin Commission was a bipartisan, one-hand pack where two senators hired five economists to underestimate the CPI by one point one percentage points to shave a trillion dollars off of Social Security for a decade. So in other words, you just fudge. You have a gauge that's supposed to be a technical thing that you use for measurement, and then somebody indexes tax brackets and Social Security, which is almost all that the government does other than defense. And suddenly people realize, "Oh, we can touch the third rail by raising taxes and slashing benefits just by getting a one point one percent overstatement." I brought this up. The world knows it. But because I don't sit in an economics chair or a physics chair or a university chair, um, I'm not serious. And what the claim is, is why won't we put the serious people in serious chairs? Why are you guys all quiet? Why are you so silent? Where is Serge Lang? Who knows who Serge Lang is? Okay. Serge Lang is a dangerous motherfucker. We need a Serge Lang in your physics department. We need people who don't know when to shut up. We don't have any of them because we've got these deans and administrators and these hurdles and these grants and all of these things. We're, we're getting smaller and smaller and smaller. And there's some point in which you say, "Look, we are cowboys. We are Americans. We are free. We can say what we want, and we're not gonna mouth your lies. Give us our money. Go away. Let us do our job." And the country loves this. Unfortunately, and I'm gonna be honest about it, the right wing has a point. It's not as good as they think it is. But the amount of group think inside the university trying to get positive social outcomes by changing the norm, it's too high! don't even think it should be zero. If you, if you catch me on the Sam Harris program, one of the first podcasts I did, I said, "I don't only care about truth, I also care about meaning, human fitness, and grace." Right? We trade off between various things that are important. Truth is of paramount importance to what we do in science. It's also important to remember that this is about meaning. It's about fitness. If we, if we weaponize anthrax and, and spread the recipe, that's not necessarily good because it's science. And grace is important to the earlier point about transgender people. Decency.

02:14:27,866 --> 02:14:29,146 [speaker_3]

Can you talk to me, William?

02:14:29,326 --> 02:15:05,996 [speaker_13]

Yeah. I mean, perhaps this is another thing you'd like to sort of attack the fiction. One sort of-- Perhaps to bring it back to, uh, pointing to physics. One sort of story I've heard a lot about kind of why physics is, uh, funding for high energy theoretical physics is at least kind of on the rocks is, and this is partly, partly our fault, partly not, is the fact that, um, a lot of the experimental work in high energy physics, namely on like CERN, kind of hasn't produced some of the results we thought we were gonna get, and we kind of overhyped this, kind of overplaying our cards a bit. I guess I'd be curious to know kind of what your-

02:15:06,066 --> 02:15:06,806 [speaker_0]

Disagree.

02:15:12,446 --> 02:15:12,576 [speaker_13]

Okay.

02:15:12,576 --> 02:19:05,486 [speaker_0]

Okay. A lot of you guys are bad at science. You got results on superpartners that are extraordinary. They're not where they're supposed to be. This is really interesting news. That's a negative result. You won. You found something out about the world that we didn't know. Okay. Did you ever ask who told us that there would be no superpartners? Why did you tell us that? What's more, do you ever ask the question, is there anybody who's a supersymmetry proponent who said we won't find super, superpartners? You suck at physics. That's your problem. Your problem is, is that you got great negative results. It's super interesting, and you don't even do basic science, which is to say, "Okay, who bet what on what?" You're so worried about your gatekeeping, that somebody's gonna give a talk that's gonna have some pseudoscience in it, that you don't even ask basic questions. And so this is m- this is part of my frustration with this. You also have a Ken Wilson problem. Nobody talks about it. Is there anything wrong with what Ken Wilson found? Nothing. Ken Wilson is one of the great geniuses of the field. He is-- Ken Wilson is the Alan Holdsworth of theoretical physics. There are no people outside theoretical physics who can write a Ken Wilson post and run with it. Right? It is not always the case that the right way to titrate our understanding is through energy levels. Scale has infected the minds of people who follow Ken Wilson. That's not to say that it's not an important insight, but the whole point of doing that Wheel of Fortune thing is, is that very often it's a bad way of cross-sectioning. If you think about additive pr- additive manufacturing through three D printing, why would you take something like a moz and do a, an individual little dot, and go round and round and, and cross-section it? It's like a-- It's a bad way of thinking in some cases. So my claim is, is that, one, the experimentalist theory interface mediated by misinterpretation of Ken Wilson is a disaster. The inability to ask the question, were we wrong? Did anyone else get this right? We're slaves to this narrative that isn't true, which is that there are no other games in town. Servicing the narrative has cost us our credibility. And I think that that's really the problem, is, is that so many people bought into the idea that you can only see more through more energy, and that there is nothing. We, we've all tried it with the, with the particles that we have, and nobody has any new ideas. Totally, totally wrong. But we've allowed that narrative because somehow it flatters those who failed, and we've got to stop flattering those who failed. I have no person on earth who I revere as much for contributions alive today as Edward Witten. Edward Witten drove theoretical physics off a cliff almost single-handedly with Leonard Susskind, Andy Strominger, with David Gross, with a few other people. It's not an anti-Ed poll. Ed's the greatest genius I've ever met. He did more for quantum field theory than anybody alive. You have to be able to do science and not just service the flattering narrative of your leaders.

02:19:09,446 --> 02:19:11,366 [speaker_3]

You had a good one.

02:19:11,386 --> 02:19:25,106 [speaker_13]

Hey, Dan. What, what narrative do I tell my clients to where they wanna come invest in physics instead of bundling up a dentist office in the Midwest again or buying their tenth yacht? Like, how, how do I get them-- How do I get money here?

02:19:25,666 --> 02:23:26,762 [speaker_0]

Great question. Philanthropy has changed its name to giving, which I find irritating, but fine. We talk about the giving community. We don't talk about the taking community. When I went on a trip through the-- across the Great Himalayan Range when I was twenty or twenty-one, and I went into these remote valleys, and people-- And say, "Could you please do surgery on my hand?" Who had a giant tumor on her face. I was like, "Why?" "Well, because you're white." Okay. Well, some white doctor had been there years ago, and they had a memory that somebody like me, um, had done something positive. So I thought a lot about that when I got home, and I found all these people doing cleft lip and glaucoma work in remote communities. One of the most beautiful things I ever found was that people would go back years later and they would find all these children the doctors made. The giving community can be honored and flattered for their taking a risk on you in a non-transactional fashion. Most of you need to unlearn the word transactional. Wouldn't you think it a good idea if we named some stuff after the people who believe in us so that they have a durable opportunity to go into the future? We are the keepers of immortality, and I'm gonna recommend everybody take a look at a picture by Jan van Eyck called Giovanni Arnolfini and his wife. Giovanni Arnolfini was otherwise forgettable merchant. I think it's of Italian extraction in Holland or something. But he picked Jan van Eyck to do his portrait with his pregnant wife. And Jan van Eyck wrote himself into a mirror that's in the background, so you see the artist painting this couple. And in Dutch, I think it says, "Jan van Eyck was here." [foreign language] Right? But the point is, is that they went into immortality together, Jan van Eyck and Giovanni Arnolfini, right? My feeling is, why not ask the question-- The gentleman before who asked, you know, "Why don't you just open your checkbook?" He didn't say the right question, which is, "Can you tell me what, what would be meaningful for you as a gift that could come from our community?" And what you should think of it is that you're rich. You're extraordinarily rich. You're rich in a different currency. It's very hard to get at. And my claim is act with it. Act rich. Think of yourselves as a giving community who can bestow something. One thing that's really amazing is getting private seminars from the world's smartest people. You know? Uh, nobody wants a twelfth yacht. Uh, what they want is they want something special. And I'll just say I have no idea what, what you've formed as a belief about Peter Thiel, who's a clo- a close friend and somebody I know quite well and somebody who I fight with too from Yale. Peter cares about two things in life. He says ideas and winning. And the more he wins, the more he can gather people to share ideas. And the more ideas he gets, the more he can figure out how to win, and he just keeps that as an infinite cycle. You may not like him. You may think he's a horrible person. That's fine. But my claim is, is that ultimately all wine tastes the same when you're rich enough, and most experiences just blend into each other. The thing that doesn't is exploration and the new and the frontier. If you guys can actually make progress on the frontiers, the right question to ask is, "What can we do for you in return?" And I think you'll find that that works.

02:23:32,902 --> 02:23:50,852 [speaker_22]

So, uh, [clears throat] suppose you've laid out a case here. So supposing I follow it, what can one do as a PhD student when resources are so scarce and potentially rocking the boat can mean-

02:23:50,892 --> 02:23:53,502 [speaker_0]

How good are you?

02:23:53,562 --> 02:23:54,242 [speaker_22]

I'm good enough.

02:23:55,702 --> 02:26:07,018 [speaker_0]

Bad answer. How good are you? It's an uncomfortable question, but let me tell you what I'm hoping you're saying internally. Better than you. If you're not confident that you're the, that you're the one, it's not really clearly a great way to use a life. It's so hard to do this. Um, so many of you were taught modesty, and my belief is that modesty and epistemic humility are really important virtues. But going back to Ecclesiastes, there's a time and purpose under heaven for everything. There's a time for arrogance. If you think you're really good and you have an insight-- Um, you know, for me, when I saw the particle spectrum of the first generation of Fermi, I just fell out of my chair. Like, this is unbelievable. I thought everybody must understand what this is saying, and nobody did. And then I found other things, and nobody knew them. And so I just became convinced, "Oh, they're all wrong and I'm right." Now, has it been a good life? Sort of. Sort of not. Um, you should have a guess about what the universe is that's sharp enough that you can get slapped in the face and it can blow up on you. If you don't have a guess about the universe, my feeling is go into AI. Go into something else. This is the hardest thing the human mind has ever to do. It's the most consequential. But you don't have to slay that if you just find it sort of interesting. And that's the thing, which is if I give advice to musicians, the odds of making it in music are essentially zero. But that's not even true. You, you can tell early who has a chance of making it with high probability. Those people have much better odds, right? And so what I'm claiming is, is that somebody should level with you and tell you just how good you are. By the way, just to get a read, we've been at this a while, very different questions. Is this interesting to you guys?

02:26:08,278 --> 02:26:11,058 [speaker_3]

People are still here. They take this as a sign.

02:26:11,098 --> 02:26:13,098 [speaker_0]

Yeah, but I'm, I'm counting on the sample bias.

02:26:13,318 --> 02:26:16,297 [speaker_3]

We don't, we don't force them to stay this long.

02:26:16,378 --> 02:26:16,718 [speaker_0]

Okay.

02:26:18,378 --> 02:26:26,228 [speaker_3]

We, we are-- We-- But fortunately, we seem to have the room normally before people are supposed to be out. We are an hour over. Nobody comes in and boots us out, I assume.

02:26:26,228 --> 02:26:27,618 [speaker_0]

I would have brought a bat. Okay.

02:26:29,118 --> 02:27:52,998 [speaker_4]

Can I build on this just a little bit? Um, so I have a unique perspective. I'm a successful tech entrepreneur. Um, also I hire and employ about seven or eight PhD physicists. And what I would say is that the model of success, if you wanna be successful in this room-- 'Cause when I, when I put a job out, I have beginning level salespeople that make more than my PhD physicist. Beginning, right out of high school, they make a hundred thousand. Physicists make seventy-five or eighty or, you know, a hundred. But not very much money. So what I'd first say is when, when the person over here is combating the thing about the quid pro quo in terms of what does he want out of it, I think what he wants is he wants to get agency. Agency is where you get successful. So that's the first part, agency. And the second part is I feel like this entire

room is geared as he is a failure. He's the one that's the outcast. He's actually the suc-success. He's the model. If you wanna be successful in physics, that's the model. Not-- Like, you guys are great, but not the professors. This is-- This institution is like a paper clip, paper clip factory. It just makes paper clips a lot and a lot and a lot. And so the problem is, it's just, it's this flying man problem. I don't know about the immigration kind of thing, but it's too many PhDs. There's too many MBAs out there, and you have to be different. You have to be unique, and yes, you have to be the best if you wanna even have a chance of making it as a physicist. I'm a mathematician. You're a mathematician. Same thing. Yes. [laughs]

02:27:53,068 --> 02:27:53,918 [speaker_0]

What, what sub-area?

02:27:54,518 --> 02:27:55,118 [speaker_4]

Algebraic geometry.

02:27:55,138 --> 02:27:56,198 [speaker_0]

Oh, yeah, yeah, yeah. I remember you.

02:27:56,378 --> 02:28:02,168 [speaker_4]

Yeah. That's what success looks like. It's possible.

02:28:02,198 --> 02:30:34,318 [speaker_0]

But just as an example, imagine you're Grothendieck and you think, "Oh my gosh, um, people aren't realizing that they shouldn't deal with prime i-- with maximal ideals. They should deal with prime ideals. And oh my God, I have this whole structure, and I can make number theory look like..." You know, it's one idea. It's one beautiful, simple idea. Even the great Grothendieck can be reduced to something like that. What I'm trying to say is, if you don't have a bunch of hunches, if you're sitting around saying, "Oh, I'm trying to look for a problem," and you're twenty-seven, something went wrong. You should have a guess about the universe, and you should be-- just be wrong. Cut your losses. It's a beautiful life, you know? And you'll always have this to go back to, but my feeling is, is that you should have something where you might be right and everyone else on Earth might be wrong, and it's not that rare. It's not an arrogance. You know, I mean, I think maybe that's kind of-- Uh, you wanna read something great? Jim Watson, he wrote this thing called Succeeding in Science: Rules of Thumb. And it's completely subversive because what he said was, "There's a difference between great science and good science. They're not the same. And if you wanna do great science, here's what you should do. I don't know how to do good science." And if you read The Double Helix, OMG, that is one of the great works of literature. And there's like this moment where they do this triple helix where the sugar phosphate backbone from the inside of the nucleotides is sticking out, and they call in all of their colleagues, and they just laugh at Watson and Crick. And there's another meeting room, Watson and Crick are dealing with Chargaff, who says, "You know nothing about biochemistry." Right? And then Chargaff has to eat shit and writes this book called Heraclitean Fire. Is that appealing to you? Do you get off on the idea that you could be right and the other guy who's just so effing arrogant and snooty is gonna eat crap the rest of humanity because he, he shorted you? And by the way, he was a genius. He was great. He's the guy who came up with the A, C, and T and G equimolar relations. My claim is you should have lots of guesses. You should be curious, and you should look at, like, your colleagues and say, "Why are you all so lifeless and brain-dead?" If that isn't you, my advice is get out of the field. There's no nice way of saying, and it's not there to be mean. It's that the most brilliant people in the world can sit here and get almost nowhere over an entire lifetime, and it's very unlikely that's gonna happen in almost anything else that you do.

02:30:37,178 --> 02:30:40,628 [speaker_4]

Guess one more question really quick.

02:30:41,118 --> 02:30:48,658 [speaker_3]

The world feels like we're at an inflection point. You're absolutely right about the urgency. If we are at this place where we can break, what is the good outcome and what is the bad outcome?

02:30:50,778 --> 02:30:56,398 [speaker_0]

All right. This is totally irresponsible. Nobody's gonna hold me re-responsible for this. I'm just telling you it's irresponsible.

02:30:56,458 --> 02:30:56,698 [speaker_4]

Mm-hmm.

02:30:58,278 --> 02:31:01,278 [speaker_0]

The good outcome is interstellar travel. It's dark chemistry.

02:31:02,038 --> 02:31:02,258 [speaker_4]

Hmm.

02:31:03,278 --> 02:32:20,750 [speaker_0]

It's multiple temporal dimensions. It's going back to making science fiction real on a regular basis. It's the idea of rescuing humanity from the Teller-Ulam design. It's... You ever seen these Turkish rabbits that glow in the dark because of green fluorescent protein? He did that. You know? The wonder that is science and its outgrowth in technology. You know, you say, "I had to get a CAT scan or a PET scan." You don't think about the radon transform. You just wanna know whether you got a tumor. The thing about the radon transform is, is you find out that you've got a malignant tumor [laughs] that you're really prolonged. This is just such a great and wondrous adventure. And the bad outcome is, is that nuclear weapons are sort of gated by the difficulty of enriching the right radioactive isotopes. What if it becomes really easy? Remember this kid who scavenged americium from smoke detectors and built a working reactor? Um, you know, I always ask people, who know, who knows the formula for gunpowder? Pause it. What's your name?

02:32:22,330 --> 02:32:22,340 [speaker_1]

Stephanie.

02:32:22,370 --> 02:32:23,890 [speaker_0]

What, what is the formula for gunpowder?

02:32:25,170 --> 02:32:27,370 [speaker_1]

Unfortunately, um, I only remember-

02:32:27,510 --> 02:32:27,890 [speaker_0]

Come on

02:32:27,910 --> 02:32:29,290 [speaker_1]

... saltpeter. Might have been mercury. No.

02:32:29,510 --> 02:32:31,090 [speaker_0]

Seventy-five percent saltpeter.

02:32:31,350 --> 02:32:31,630 [speaker_1]

Yeah, saltpeter.

02:32:31,670 --> 02:34:50,590 [speaker_0]

Ten percent sulfur, fifteen percent charcoal. Yeah. You should all know that. [laughing] So my claim is, look, the power of the ability to code, the ab-- the ability to do PCR and, and, and, and CRISPR-Cas9 and... Look, this is the most wonderful, horrible moment to be alive ever. And right now it's a sprint to the finish. And the thing I'm really excited about is the solar system is an escape room with three possible rocks, two of which are stupid. I tell this to people, um, STEM, Mars, Moon, and biodiversification. Learn to think big again. Learn to think Einstein is not right. Nima goes around saying space-time is over. Space-time is doomed. Think about what that means. You have these two singularities that Penrose proved you can't really re-remove, Hawking and Penrose. We don't live in space-time. Space-time is the map, not the territory. The dark matter, in my belief, is decoupled, and we can recouple. There's so many things that are cool. You know, again, just imagine that everything was a smart surface. This is boring because nothing happens, you know? You've forgotten-- I just said this thing to Peter, which is of course your phone is amazing. It's all that's left of your once limitless future, you know? Get acquainted with what it is that made Einstein Time's Man of the Century for the last century. That's your tradition. And if you're not excited about that, you're not at the forefront. And, and, and learn how to dream irresponsibly. That's really my sense. Imagine that we went back to those people in nineteen twenty-seven at Solvay, and we were them. I said this to my friend Ed Franklin. He has pictures of Kurt Gödel walking with Albert Einstein and all these other people. He said, "We are them. We've just forgotten ourselves." So thank you all for coming. Really appreciate it. [clapping]